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Yang et al.

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(54) **HYBRID MICRO-BUMP INTEGRATION WITH REDISTRIBUTION LAYER**

(71) Applicant: **Taiwan Semiconductor Manufacturing Co., Ltd.**, Hsinchu (TW)

(72) Inventors: **Ting-Li Yang**, Tainan (TW); **Po-Hao Tsai**, Zhongli (TW); **Yi-Wen Wu**, Xizhi (TW); **Sheng-Pin Yang**, Kaohsiung (TW); **Hao-Chun Liu**, Hsinchu (TW)

(73) Assignee: **Taiwan Semiconductor Manufacturing Co., Ltd.**, Hsinchu (TW)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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(Continued)

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H01L 23/00 (2006.01)
H01L 21/48 (2006.01)

(Continued)

(52) **U.S. Cl.**
CPC ***H01L 24/14*** (2013.01); ***H01L 21/4853*** (2013.01); ***H01L 23/49811*** (2013.01);

(Continued)

(58) **Field of Classification Search**
CPC H01L 24/14; H01L 24/10; H01L 24/17; H01L 28/60; H01L 24/742; H10D 1/692
See application file for complete search history.

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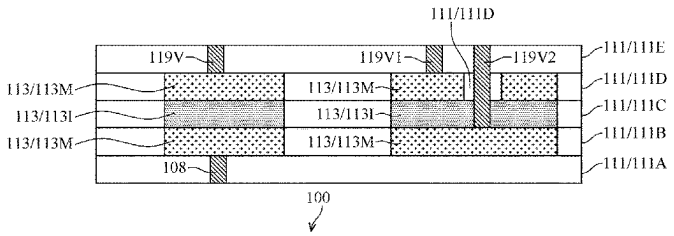
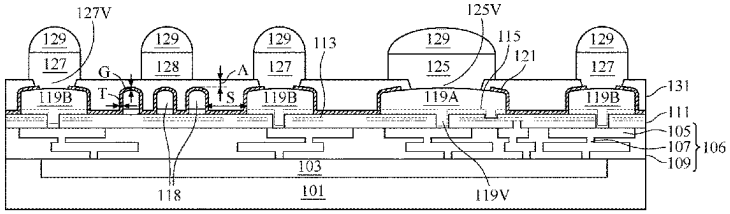
Primary Examiner — Phat X Cao
(74) *Attorney, Agent, or Firm* — Slater Matsil, LLP

(57) **ABSTRACT**

A semiconductor device includes a substrate; an interconnect structure over the substrate; a first passivation layer over the interconnect structure; a first conductive pad, a second conductive pad, and a conductive line disposed over the first passivation layer and electrically coupled to conductive features of the interconnect structure; a conformal second passivation layer over and extending along upper surfaces and sidewalls of the first conductive pad, the second conductive pad, and the conductive line; a first conductive bump and a second conductive bump over the first conductive pad and the second conductive pad, respectively, where the first conductive bump and the second conductive bump extend through the conformal second passivation layer and are electrically coupled to the first conductive pad and the second conductive pad, respectively; and a dummy bump over the conductive line, where the dummy bump is separated from the conductive line by the conformal second passivation layer.

20 Claims, 21 Drawing Sheets





Related U.S. Application Data

(60) Provisional application No. 63/139,928, filed on Jan. 21, 2021.

(51) **Int. Cl.**
H01L 23/498 (2006.01)
H01L 23/522 (2006.01)
H10D 1/68 (2025.01)

(52) **U.S. Cl.**
CPC ***H01L 23/5226*** (2013.01); ***H01L 24/10*** (2013.01); ***H01L 24/17*** (2013.01); ***H10D 1/692*** (2025.01)

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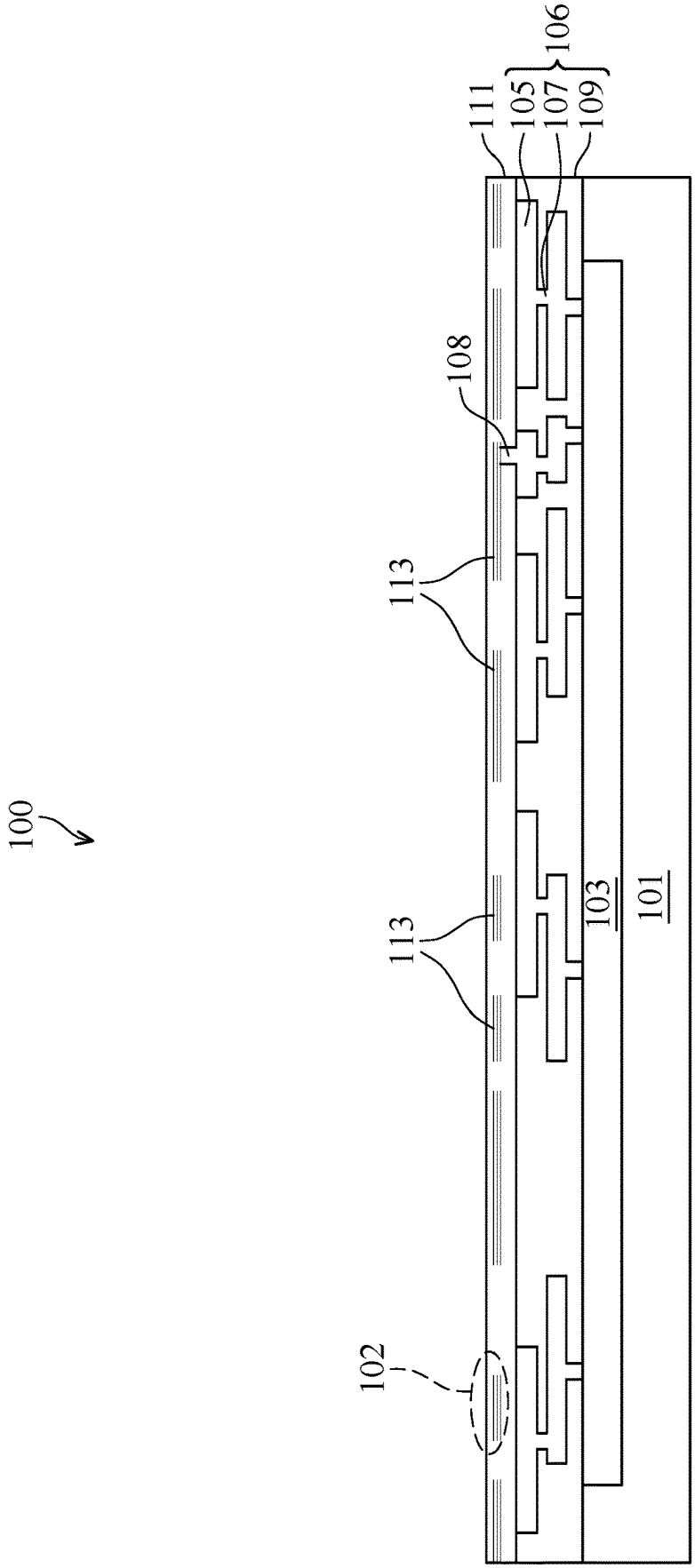
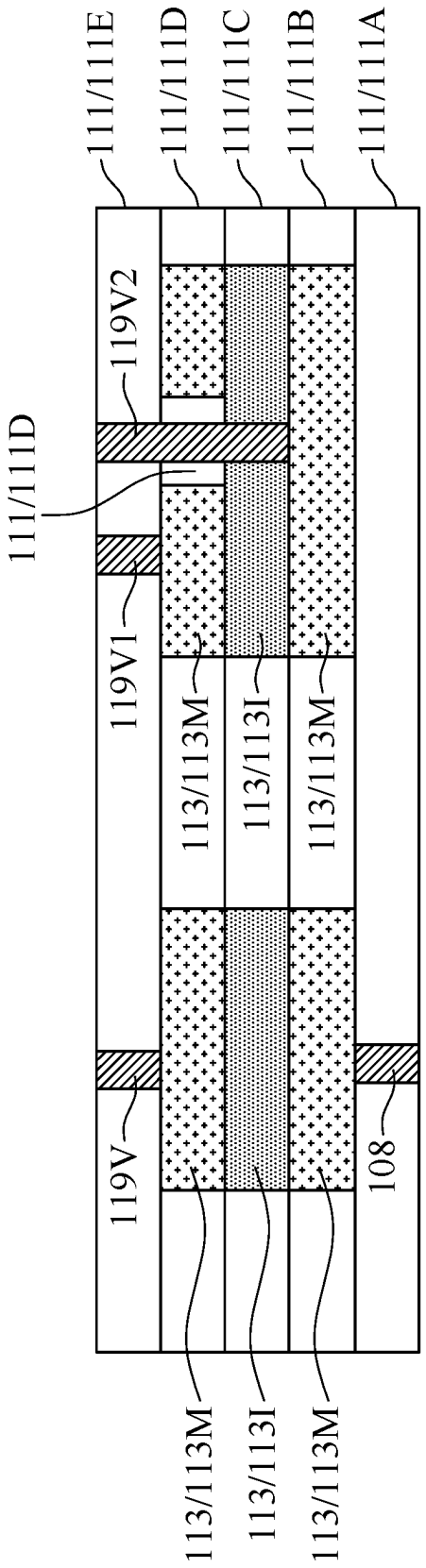


FIG. 1A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111/111D
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113/113I
113/113M
119V
108
113/113M
113/113I
113/113M

FIG. 1B

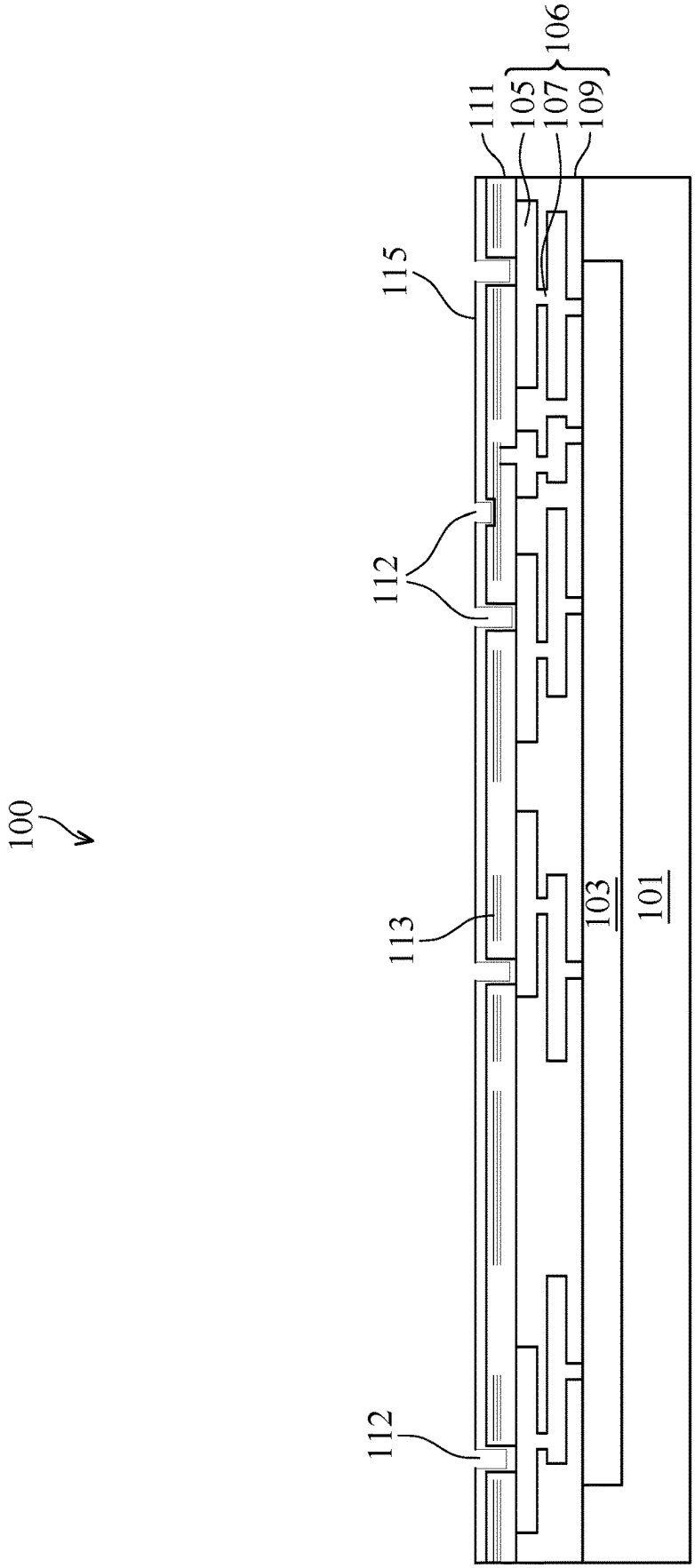


FIG. 2

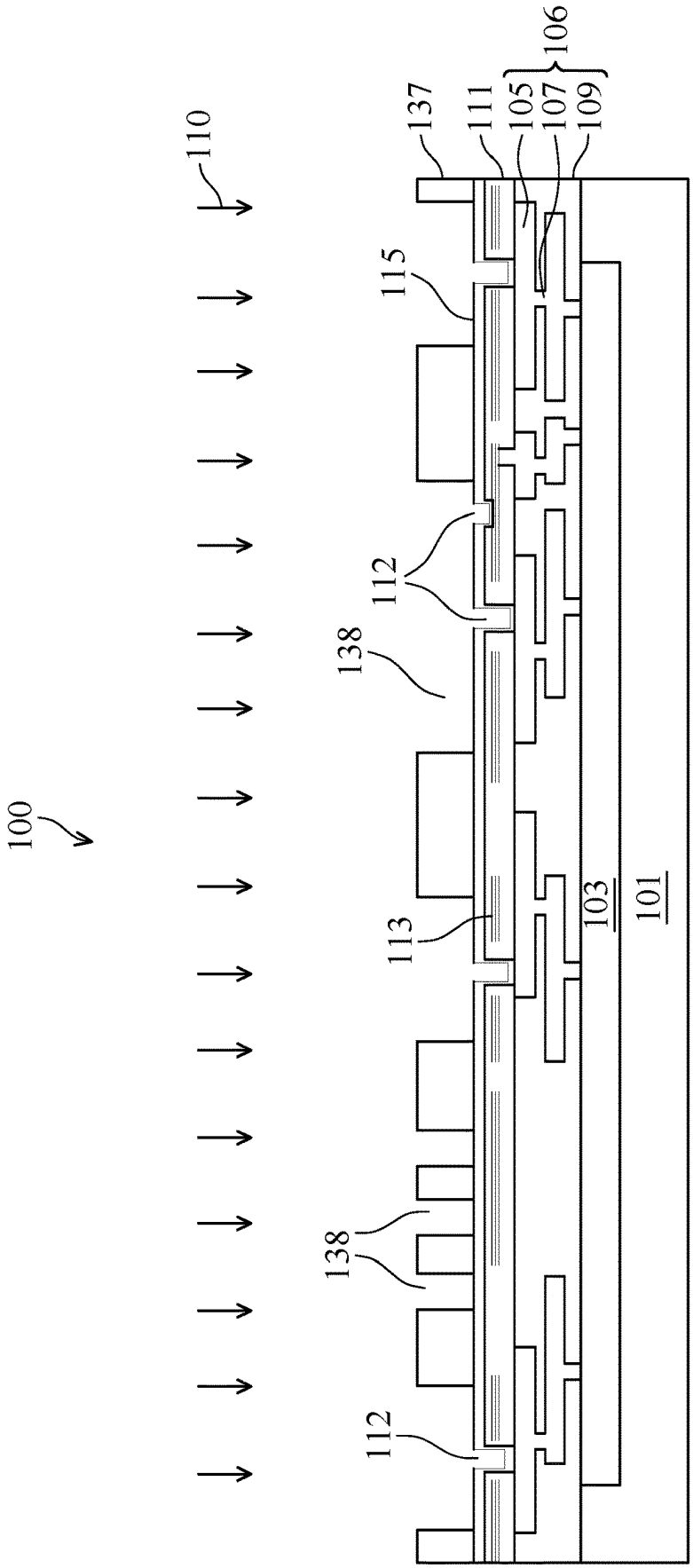
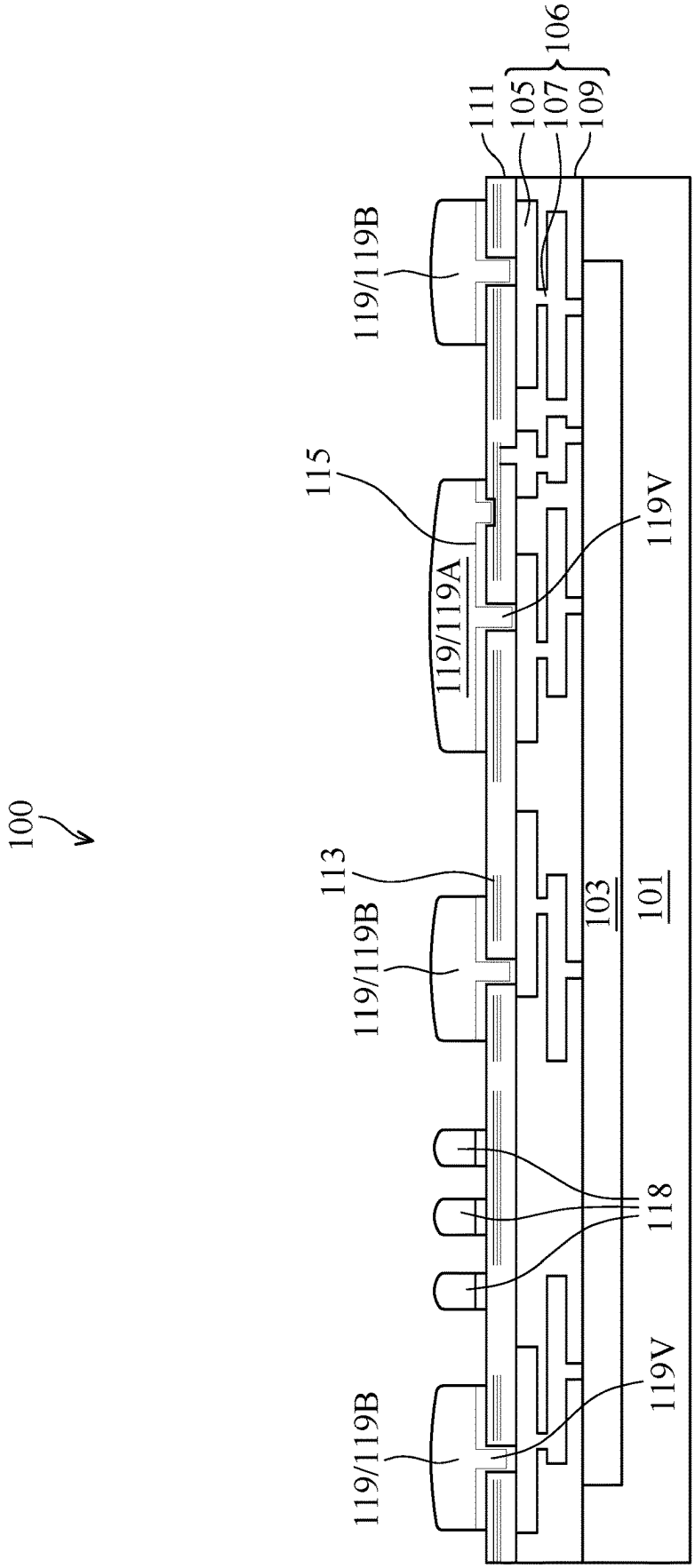


FIG. 3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119/119B
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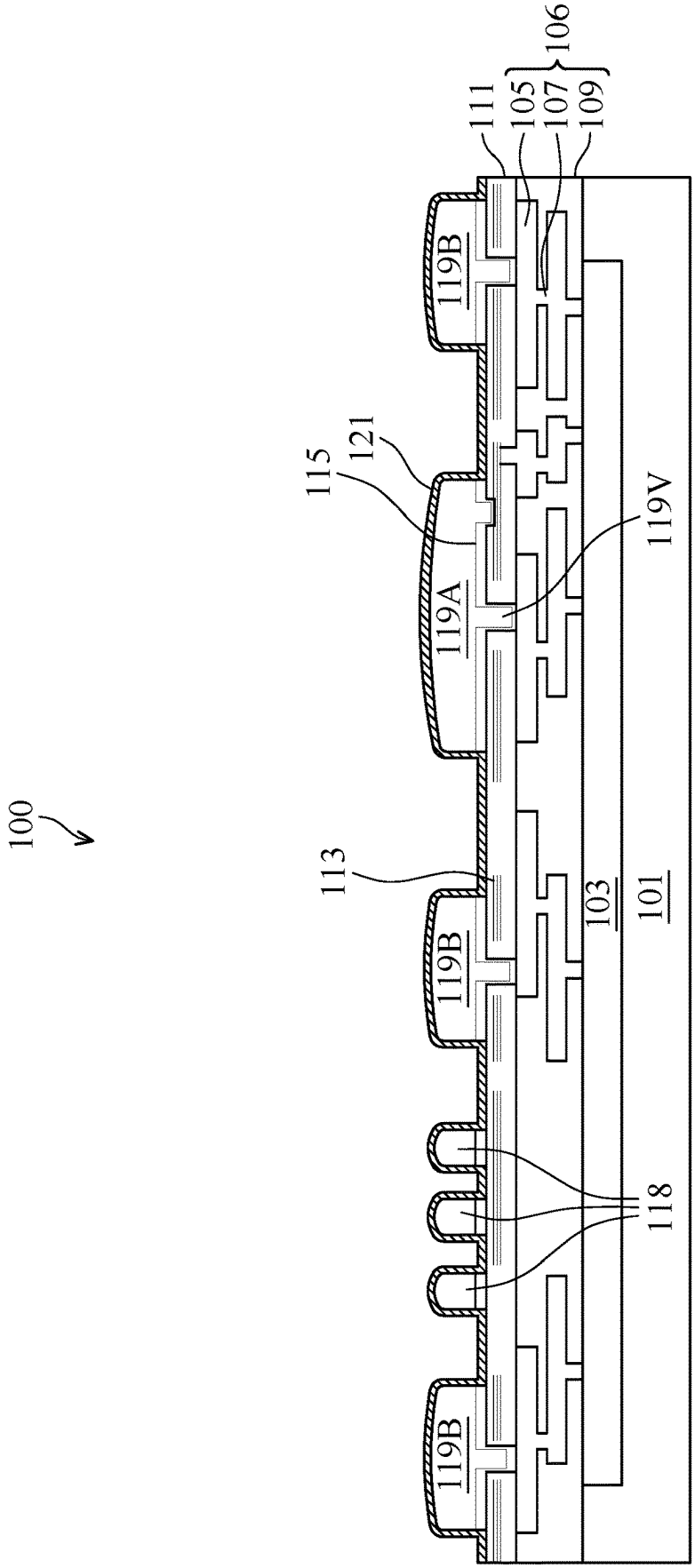
119/119B
119V

FIG. 4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119V
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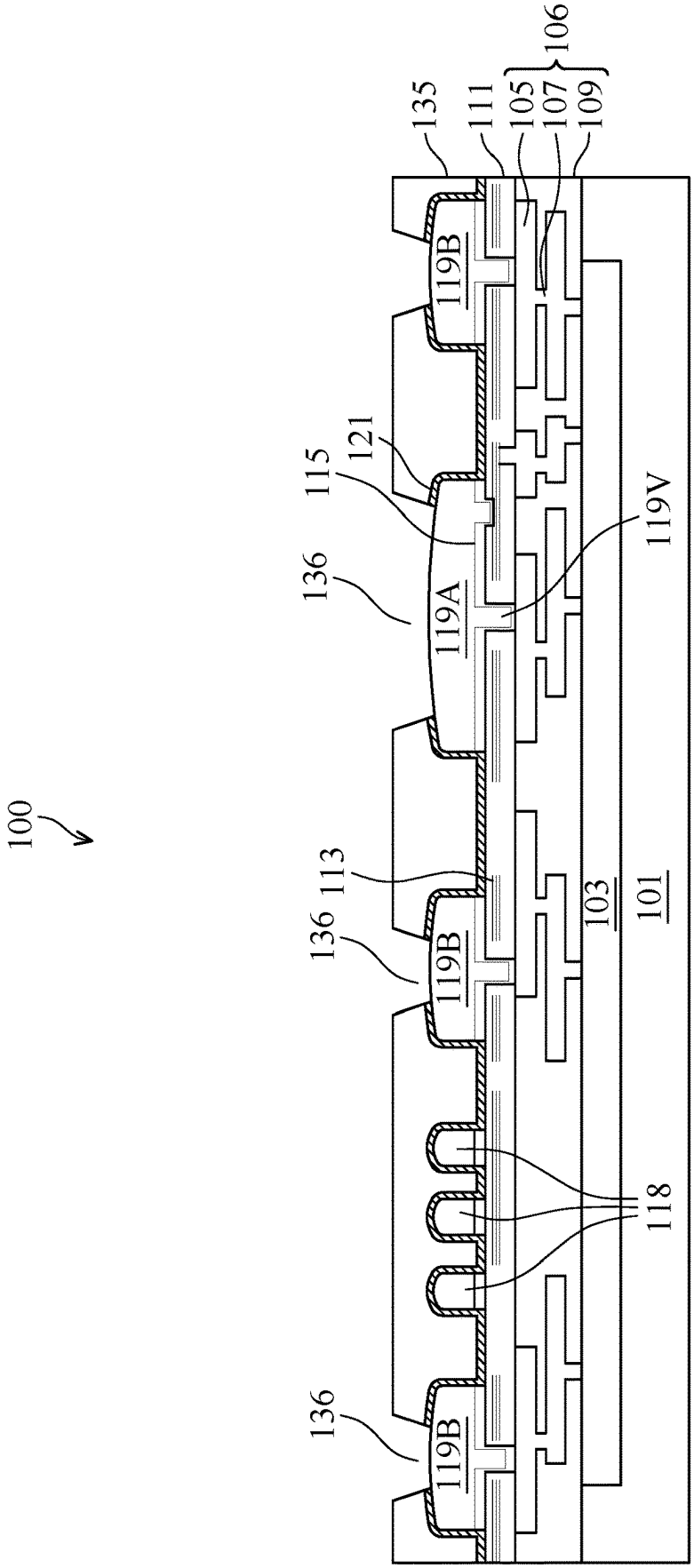
FIG. 5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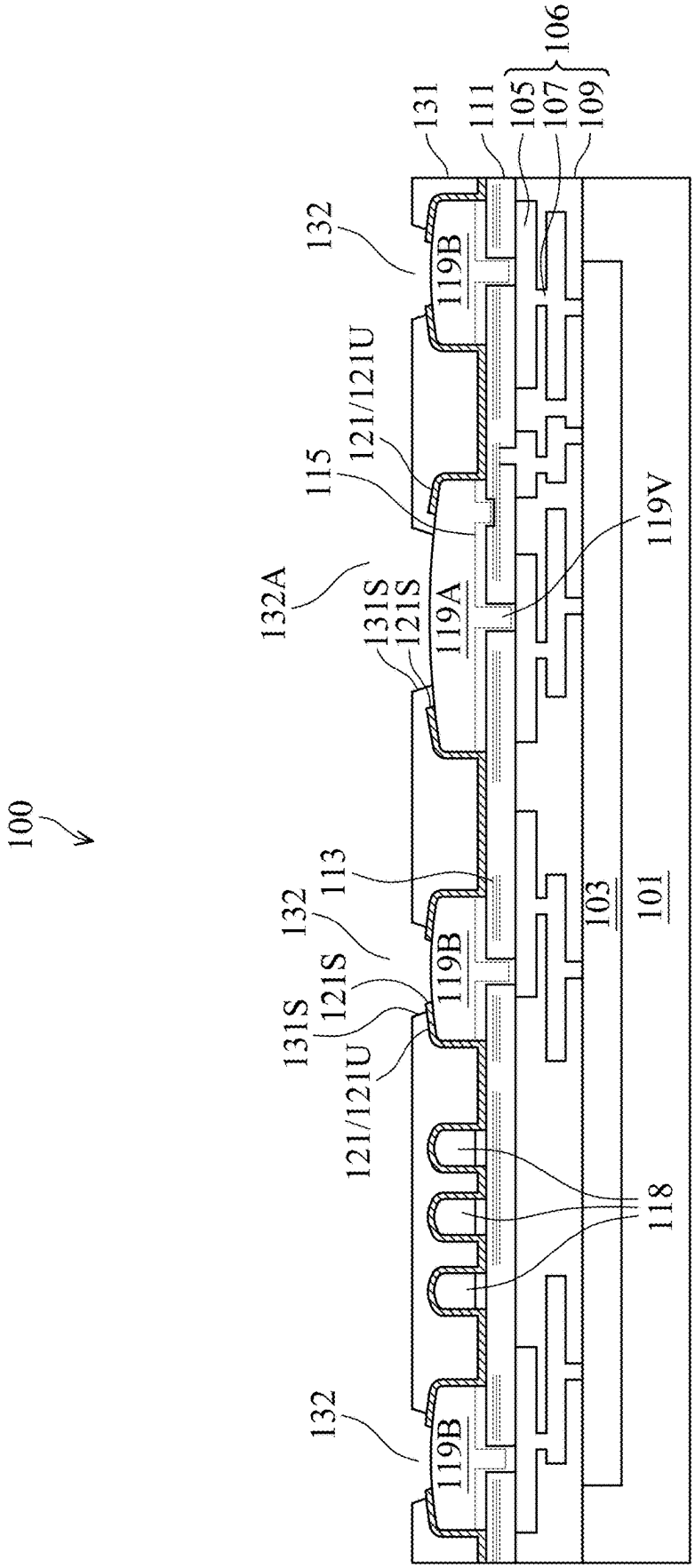
119B

FIG. 6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119V
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131S
121S
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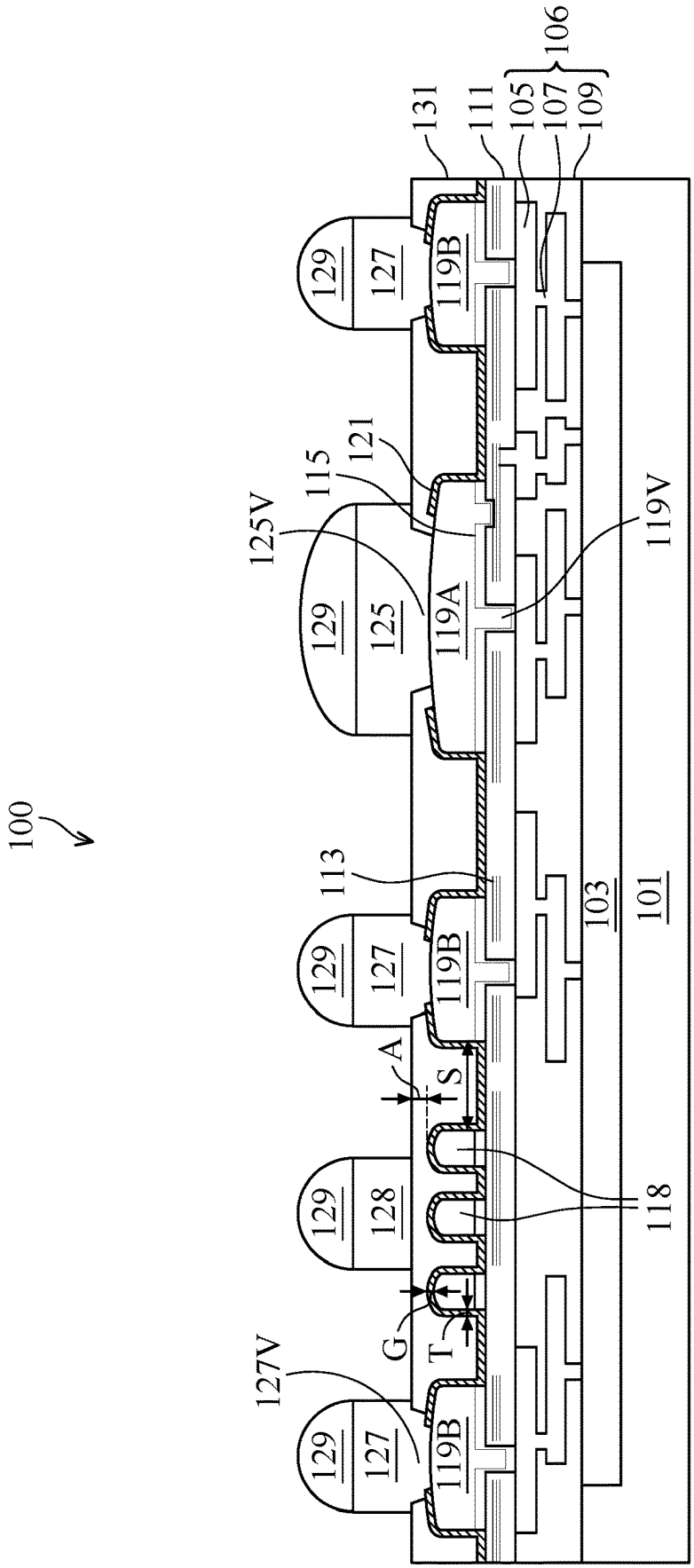
103
101
118

FIG. 7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G
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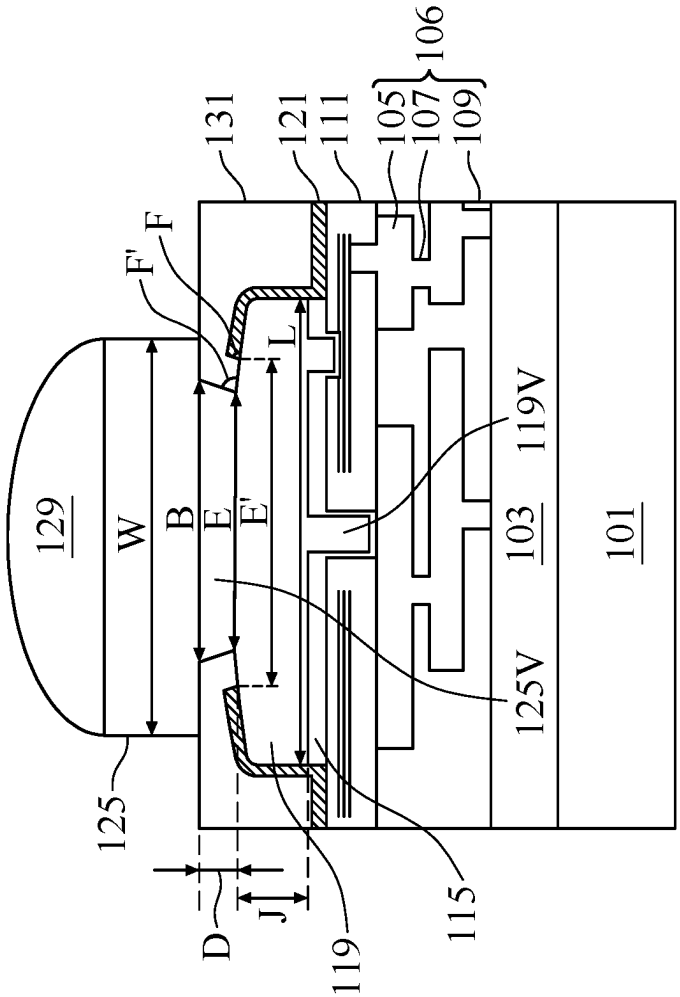
FIG. 8A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F'
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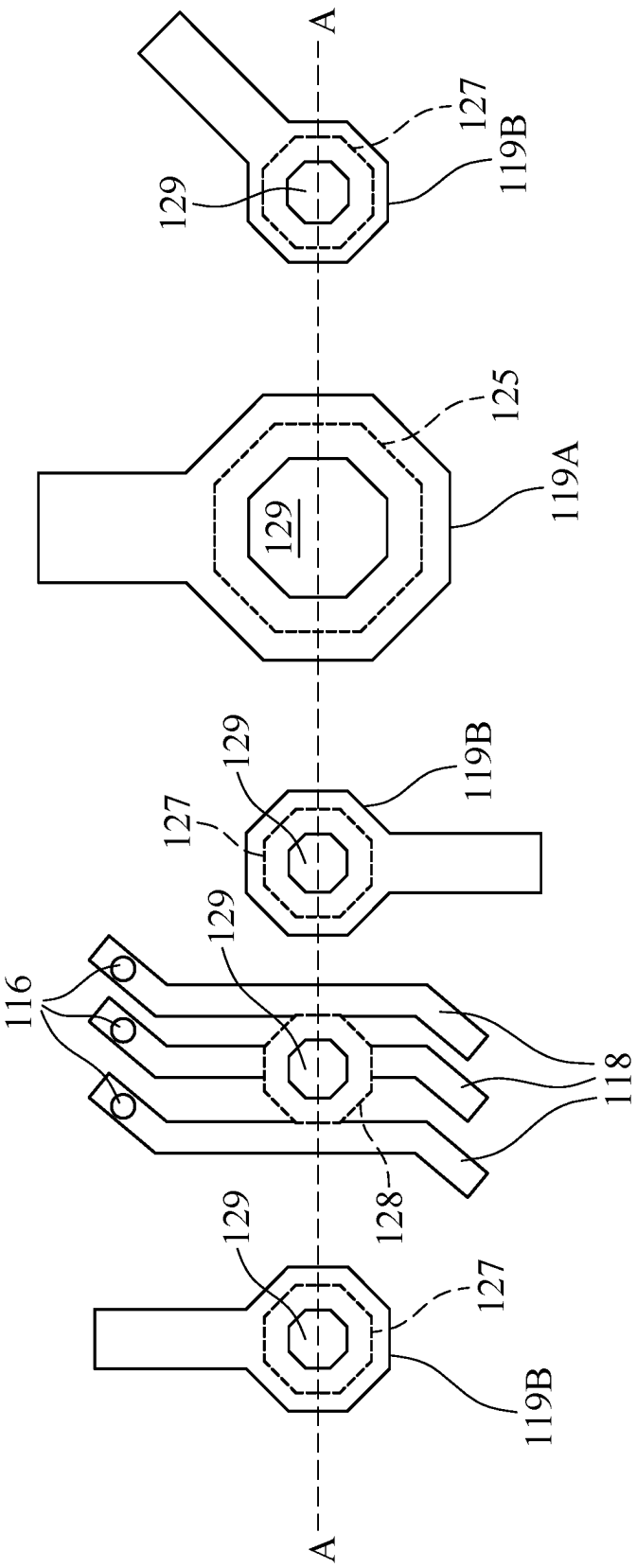
FIG. 8B


A
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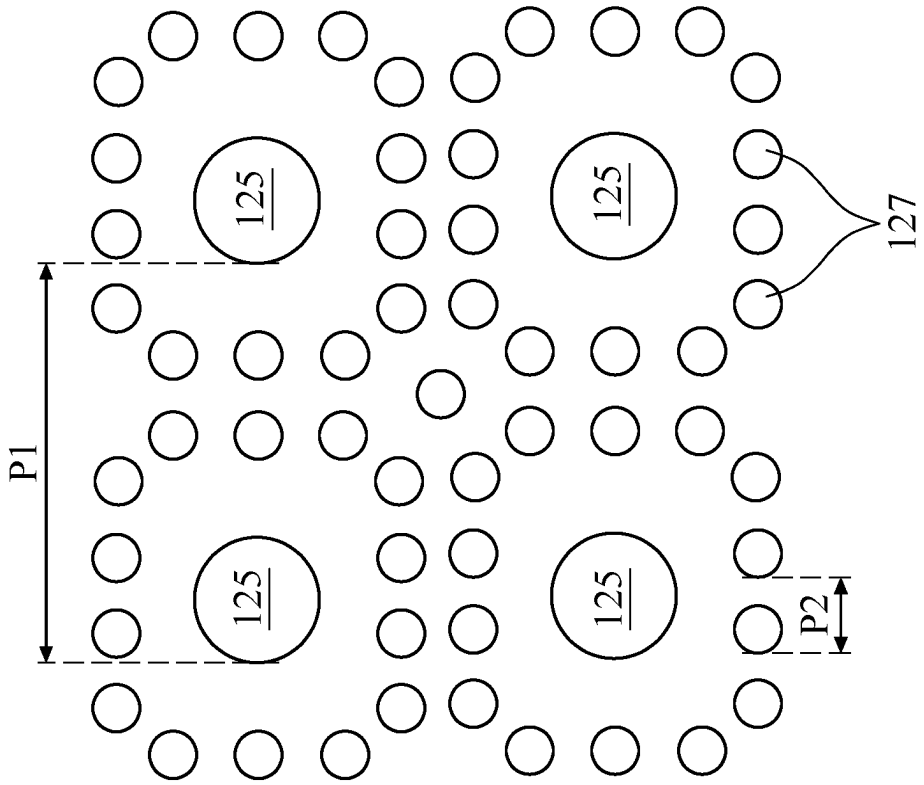
A

FIG. 8C



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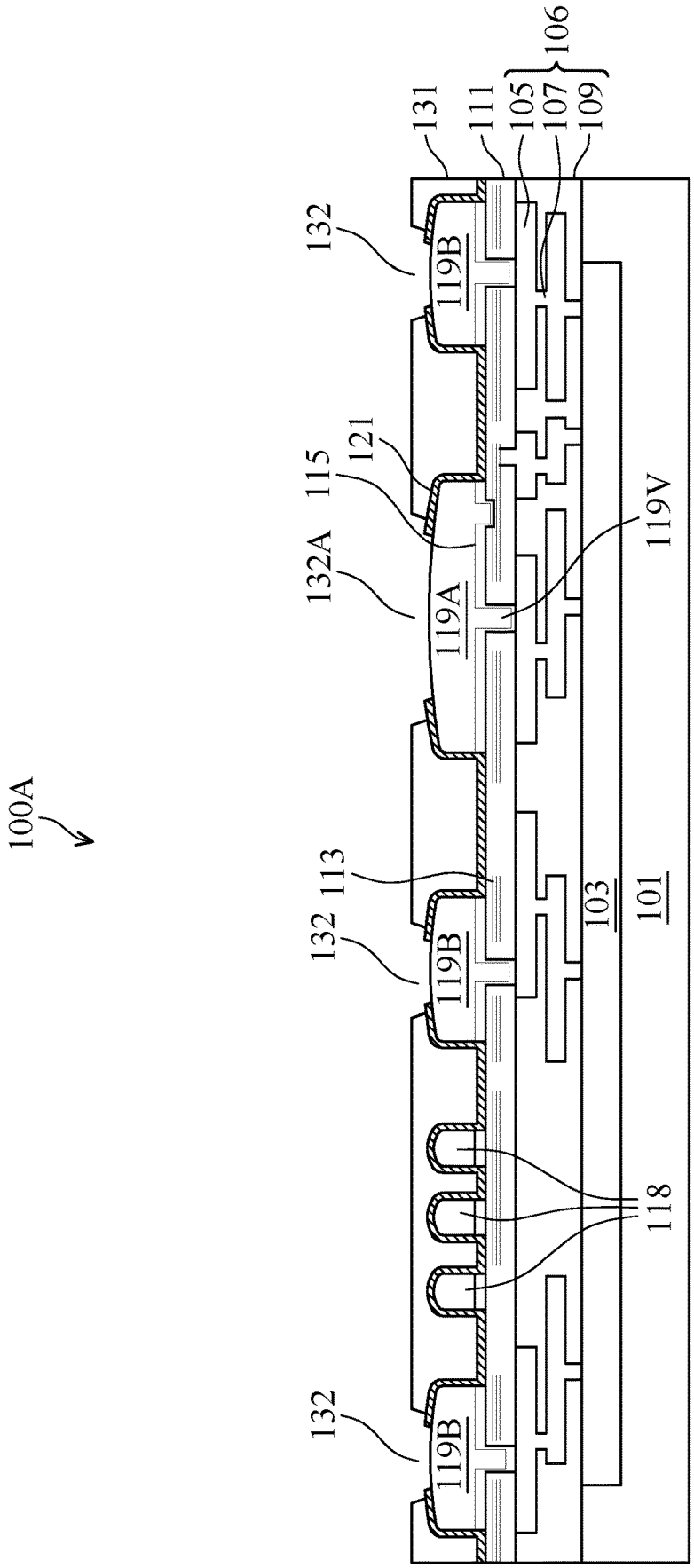
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119V
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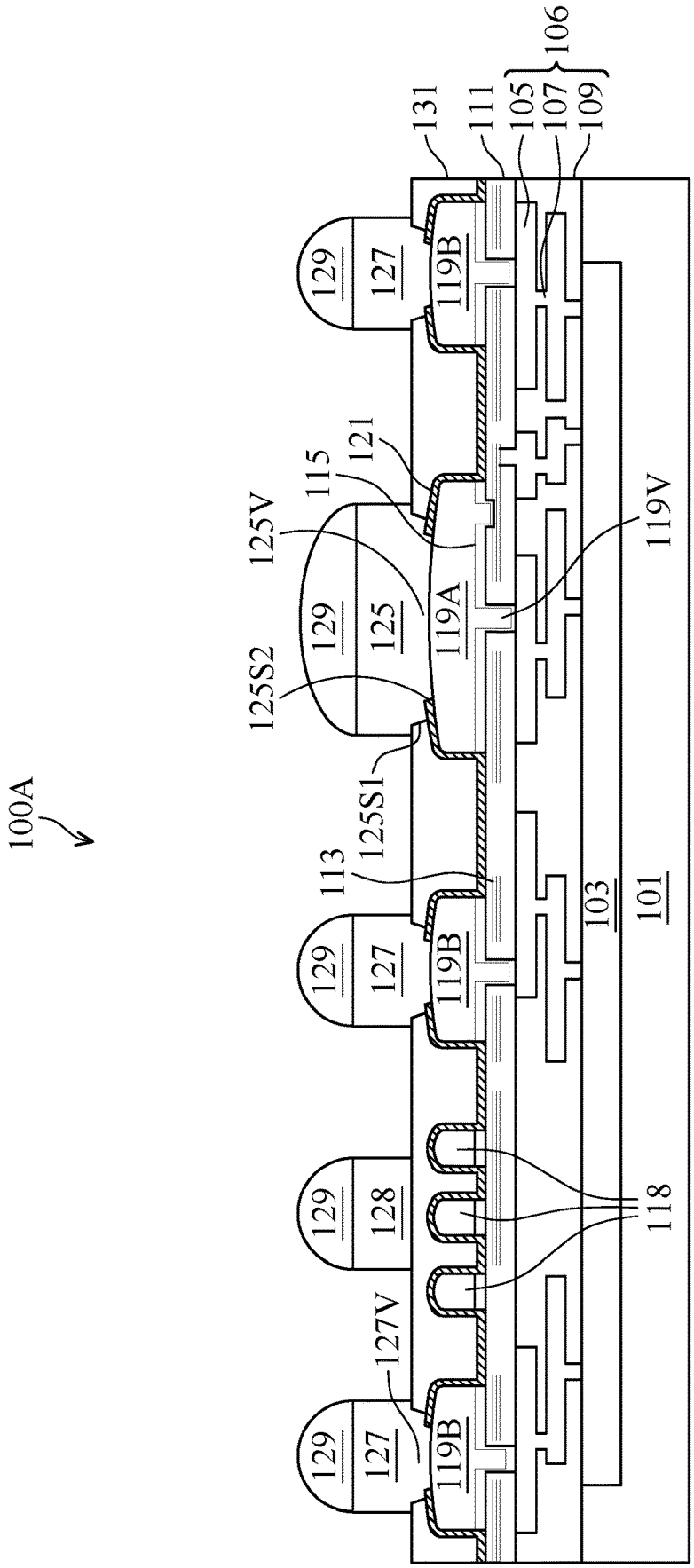
101
118
132
119B

FIG. 9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119A
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103
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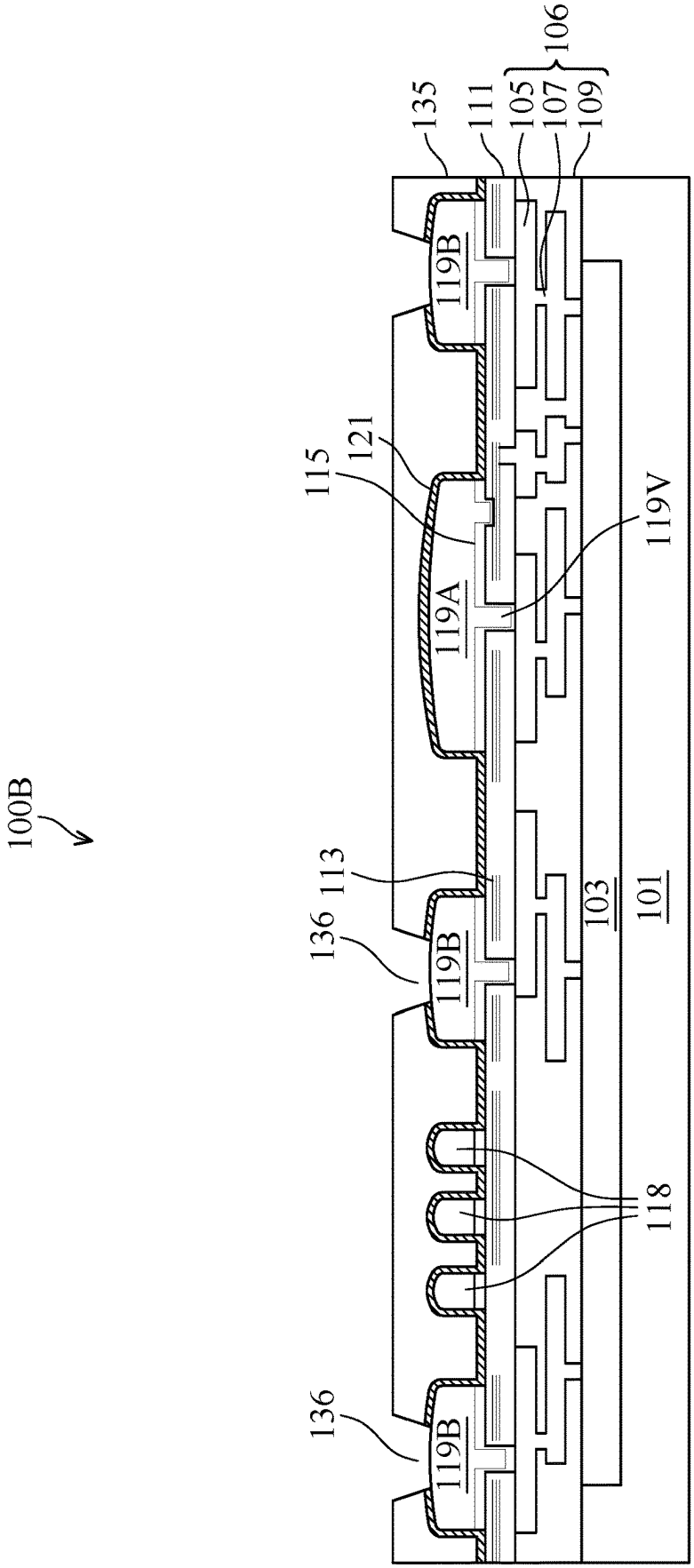
127V

FIG. 10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119B

FIG. 11

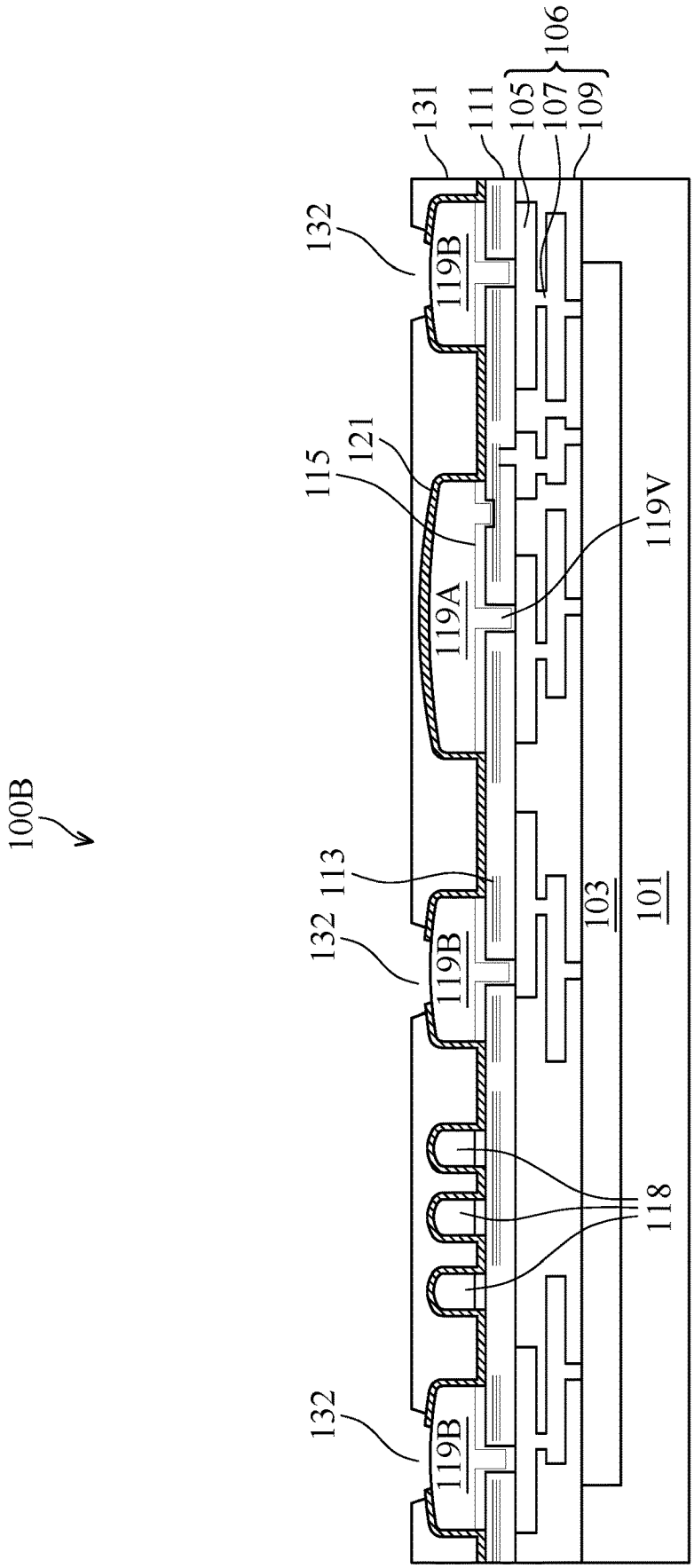
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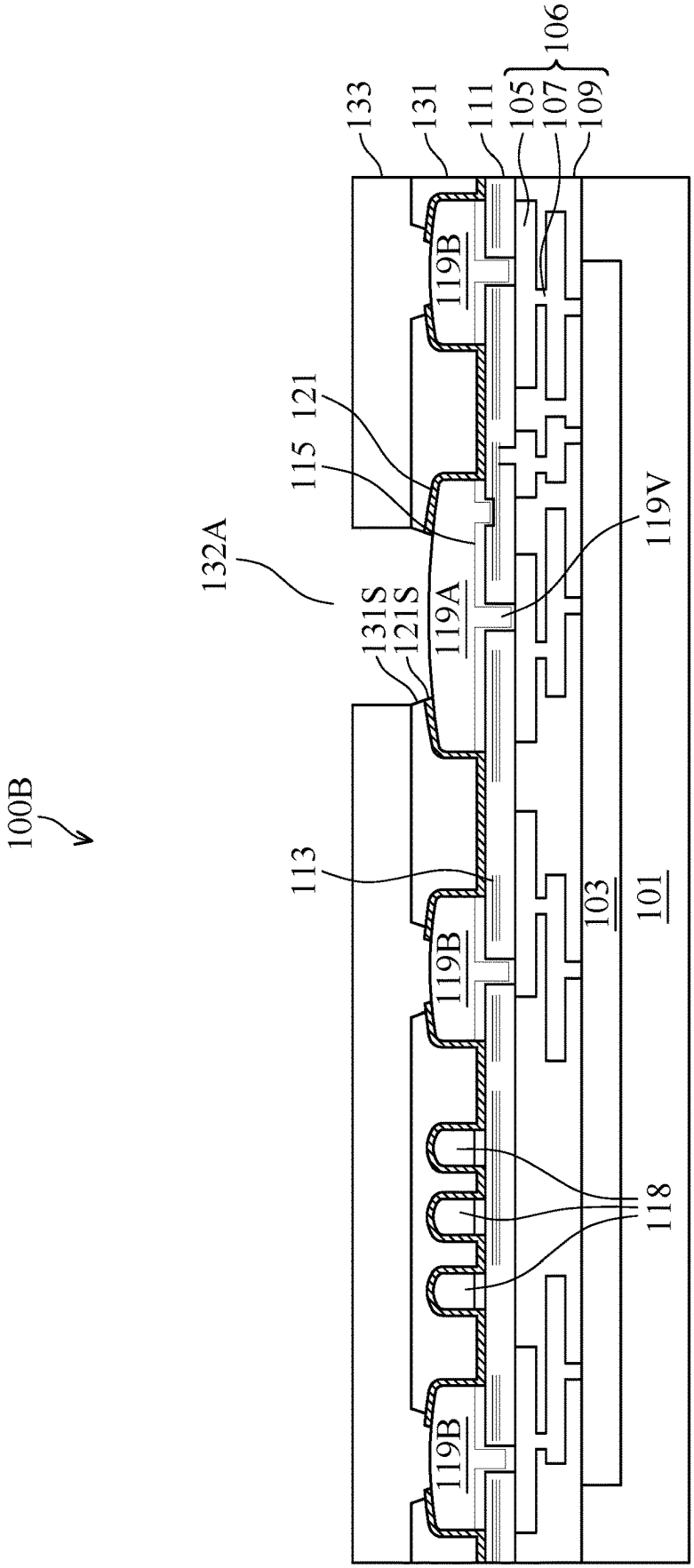
FIG. 12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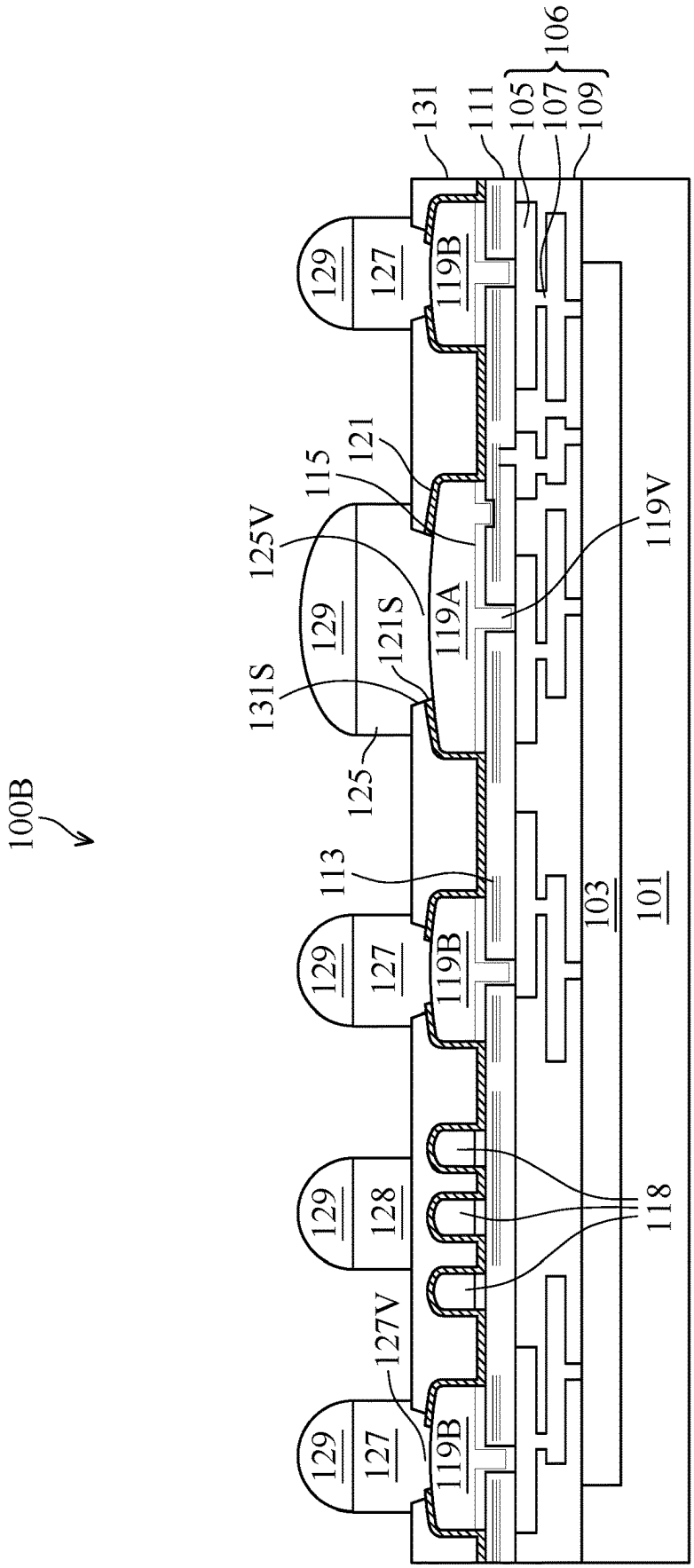
103
101
118

FIG. 13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128
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127V

FIG. 14

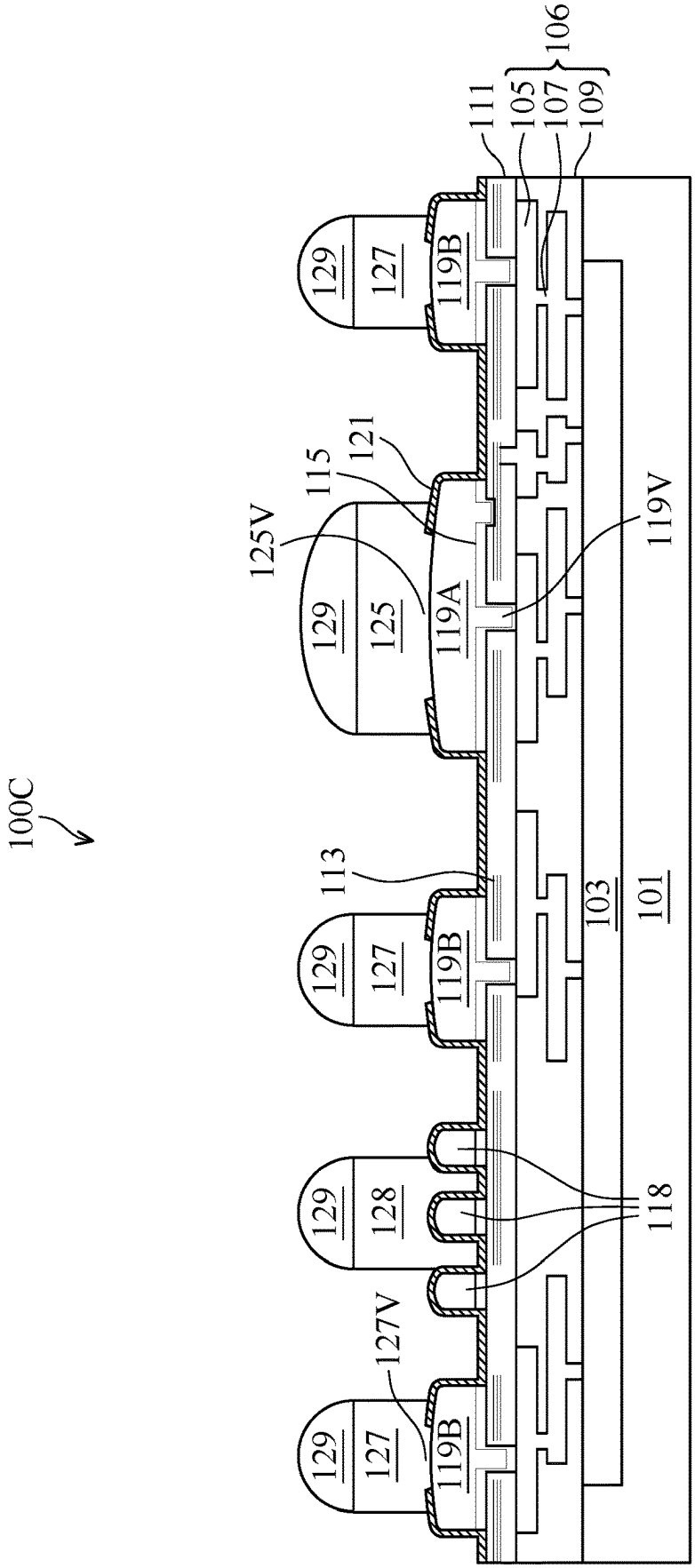
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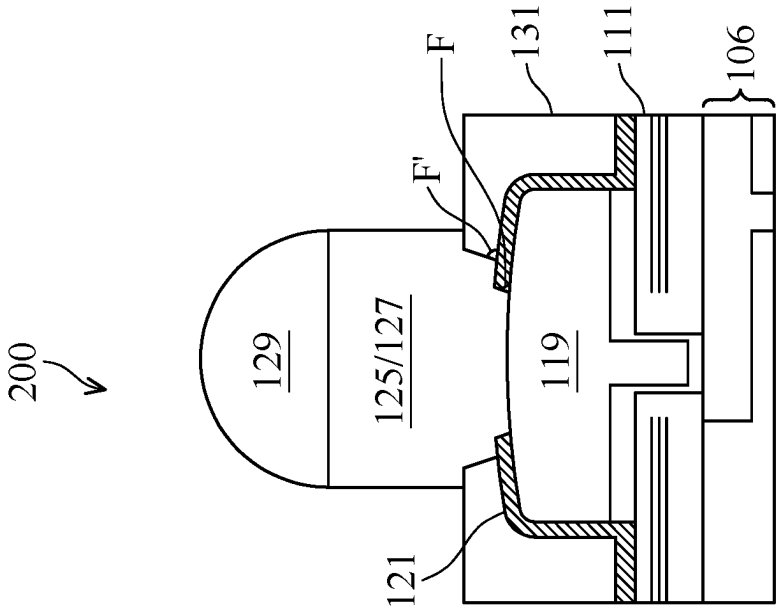
FIG. 15



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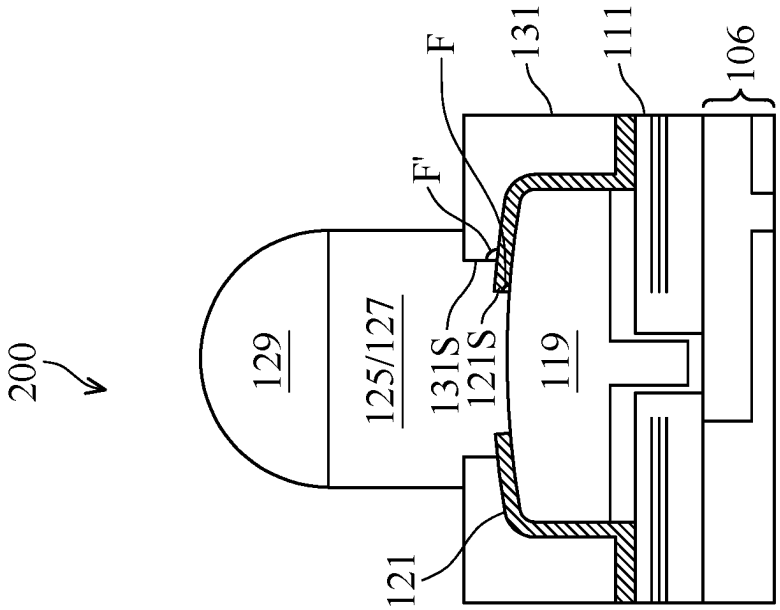

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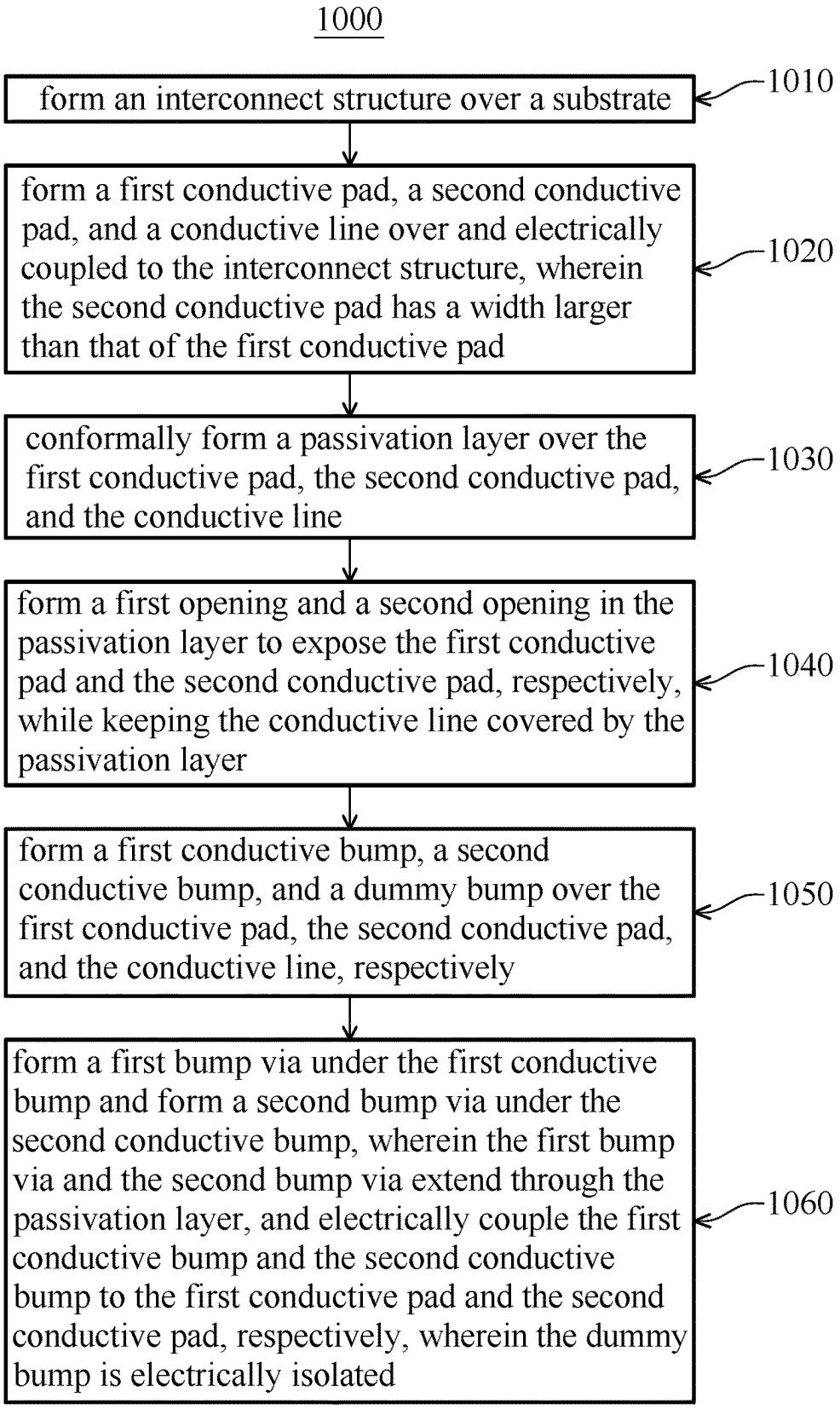
1000
form an interconnect structure over a substrate
1010
form a first conductive pad, a second conductive pad, and a conductive line over and electrically coupled to the interconnect structure, wherein the second conductive pad has a width larger than that of the first conductive pad
1020
conformally form a passivation layer over the first conductive pad, the second conductive pad, and the conductive line
1030
form a first opening and a second opening in the passivation layer to expose the first conductive pad and the second conductive pad, respectively, while keeping the conductive line covered by the passivation layer
1040
form a first conductive bump, a second conductive bump, and a dummy bump over the first conductive pad, the second conductive pad, and the conductive line, respectively
1050
form a first bump via under the first conductive bump and form a second bump via under the second conductive bump, wherein the first bump via and the second bump via extend through the passivation layer, and electrically couple the first conductive bump and the second conductive bump to the first conductive pad and the second conductive pad, respectively, wherein the dummy bump is electrically isolated
1060

FIG. 17

HYBRID MICRO-BUMP INTEGRATION WITH REDISTRIBUTION LAYER

PRIORITY CLAIM AND CROSS-REFERENCE

This application is a divisional of U.S. patent application Ser. No. 17/326,941, filed May 21, 2021 and entitled "Hybrid Micro-Bump Integration with Redistribution Layer," which claims the benefit of the U.S. Provisional Application No. 63/139,928, filed Jan. 21, 2021 and entitled "Advanced Hybrid μ-bump Integrated with Cu RDL," which applications are hereby incorporated herein by reference.

BACKGROUND

High-density integrated circuits, such as Very Large Scale Integration (VLSI) circuits, are typically formed with interconnect structures (also referred to as interconnects) serving as three-dimensional wiring line structures. The purpose of the interconnect structures is to properly connect densely packed devices together to form functional circuits. With increasing levels of integration, a parasitic capacitance effect between the metal lines of the interconnects, which leads to RC delay and cross-talk, increases correspondingly. In order to reduce the parasitic capacitance and increase the conduction speed of the interconnections, low-k dielectric materials are commonly employed to form Inter-Layer Dielectric (ILD) layers and Inter-Metal Dielectric (IMD) layers.

Metal lines and vias are formed in the IMD layers. A formation process may include forming an etch stop layer over first conductive features, and forming a low-k dielectric layer over the etch stop layer. The low-k dielectric layer and the etch stop layer are patterned to form a trench and a via opening. The trench and the via opening are then filled with a conductive material, followed by a planarization process to remove excess conductive material, so that a metal line and a via are formed. Conductive bumps, such as micro-bumps (μ-bumps) and controlled collapse chip connection bumps (C4 bumps), are formed over the interconnect structures for connection with other devices.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. **1**A, **1**B, **2**-**7**, and **8**A-**8**D illustrate cross-sectional views of a semiconductor device at various stages of manufacturing, in accordance with an embodiment.

FIGS. **9** and **10** illustrate cross-sectional views of a semiconductor device at various stages of manufacturing, in accordance with another embodiment.

FIGS. **11**-**14** illustrate cross-sectional views of a semiconductor device at various stages of manufacturing, in accordance with another embodiment.

FIG. **15** illustrates a cross-sectional view of a semiconductor device, in accordance with yet another embodiment.

FIGS. **16**A and **16**B illustrate cross-sectional views of a semiconductor device with different processing sequences, in accordance with some embodiments.

FIG. **17** illustrates a flow chart of a method of forming a semiconductor device, in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact.

Further, spatially relative terms, such as "underlying," "below," "lower," "overlying," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. Throughout the description herein, unless otherwise specified, the same or similar reference numerals in different figures refer to the same or similar element formed by a same or similar formation method using a same or similar material(s). In addition, unless otherwise specified, figures with the same numeral and different alphabets (e.g., FIG. **8**A and FIG. **8**B) illustrate different views (e.g., along different cross-sections) of the same semiconductor device at the same stage of manufacturing.

In accordance with an embodiment, a plurality of conductive pads and conductive lines are formed over and electrically coupled to an interconnect structure. A conformal passivation layer is formed over the conductive pads and the conductive lines. Conductive bumps, such as micro-bumps (μ-bumps) and controlled collapse chip connection bumps (C4 bumps), are formed over the passivation layer and electrically coupled to the underlying conductive pads. Dummy bumps are formed over the conductive lines and are isolated from the conductive lines. By forming the dummy bumps over the conductive lines, dummy conductive pads are no longer needed, and the space saved may be used to route other functional features, such as the conductive lines. In some embodiments, a dielectric layer is formed over the passivation layer before the conductive bumps are formed. The dielectric layer fills the gaps between the conductive pads and between the conductive lines, thus making it easier to form the seed layer used for forming the conductive bumps. The present disclosure allows different types of conductive bumps, such as C4 bumps and μ-bumps, to be mixed together (e.g., interposed between each other) in a same region of the semiconductor device. Such a hybrid bump scheme allows for more flexibility in the design of conductive bumps to accommodate different design requirements.

FIGS. **1**A, **1**B, **2**-**7**, and **8**A-**8**D illustrate cross-sectional views of a semiconductor device **100** at various stages of manufacturing, in accordance with an embodiment. The semiconductor device **100** may be a device wafer including active devices (e.g., transistors, or the like) and/or passive devices (e.g., capacitors, inductors, resistors, or the like). In some embodiments, the semiconductor device **100** is an interposer wafer, which may or may not include active devices and/or passive devices. In accordance with yet

another embodiment of the present disclosure, the semiconductor device **100** is a package substrate strip, which may be package substrates with cores therein or may be core-less package substrates. In subsequent discussion, a device wafer is used as an example of the semiconductor device **100**. The teaching of the present disclosure may also be applied to interposer wafers, package substrates, or other semiconductor structures, as skilled artisans readily appreciate.

As illustrated in FIG. **1**A, the semiconductor device **100** includes a semiconductor substrate **101** and electrical components **103** (e.g., active devices, passive devices) formed on or in the semiconductor substrate **101** (may also be referred to as substrate **101**). The semiconductor substrate **101** may include a semiconductor material, such as silicon, doped or undoped, or an active layer of a semiconductor-on-insulator (SOI) substrate. The semiconductor substrate **101** may include other semiconductor materials, such as germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, gallium nitride, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. Other substrates, such as multi-layered or gradient substrates, may also be used.

In the example of FIG. **1**A, electrical components **103** are formed on or in the semiconductor substrate **101**. Examples of the electrical components **103** include transistors (e.g., Complementary Metal-Oxide Semiconductor (CMOS) transistors), resistors, capacitors, diodes, and the like. The electrical components **103** may be formed using any suitable method, details are not discussed here.

In some embodiments, after the electrical components **103** are formed, an Inter-Layer Dielectric (ILD) layer is formed over the semiconductor substrate **101** and over the electrical components **103**. The ILD layer may fill spaces between gate stacks of the transistors (not shown) of the electrical components **103**. In accordance with some embodiments, the ILD layer comprises silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), fluorine-doped silicate glass (FSG), tetraethyl orthosilicate (TEOS), or the like. The ILD layer may be formed using spin coating, Flowable Chemical Vapor Deposition (FCVD), Plasma Enhanced Chemical Vapor Deposition (PECVD), Low Pressure Chemical Vapor Deposition (LPCVD), or the like.

Contact plugs are formed in the ILD layer, which contact plugs electrically couple the electrical components **103** to conductive features (e.g., metal lines, vias) of subsequently formed interconnect structures **106**. Note that in the present disclosure, unless otherwise specified, a conductive feature refers to an electrically conductive feature, and a conductive material refers to an electrically conductive material. In accordance with some embodiments, the contact plugs are formed of a conductive material such as tungsten, aluminum, copper, titanium, tantalum, titanium nitride, tantalum nitride, alloys thereof, and/or multi-layers thereof. The formation of the contact plugs may include forming contact openings in the ILD layer, forming one or more conductive material(s) in the contact openings, and performing a planarization process, such as a Chemical Mechanical Polish (CMP), to level the top surface of the contact plugs with the top surface of the ILD layer.

Still referring to FIG. **1**A, interconnect structures **106** are formed over the ILD layer and over the electrical components **103**. The interconnect structures **106** comprise a plurality of dielectric layers **109** and conductive features (e.g., metal lines, vias) formed in the dielectric layers log. In some embodiments, the interconnect structures **106** interconnect the electrical components **103** to form functional circuits of the semiconductor device **100**.

In some embodiments, each of the dielectric layers log, which may also be referred to as an Inter-Metal Dielectric (IMD) layer, is formed of a dielectric material such as silicon oxide, silicon nitride, silicon carbide, silicon oxynitride, or the like. In accordance with some embodiments, the dielectric layers **109** are formed of a low-k dielectric material having a dielectric constant (k-value) lower than 3.0, such as about 2.5, about 2.0, or even lower. The dielectric layers **109** may comprise Black Diamond (a registered trademark of Applied Materials), a carbon-containing low-k dielectric material, Hydrogen SilsesQuioxane (HSQ), MethylSilsesQuioxane (MSQ), or the like. The formation of each of the dielectric layers **109** may include depositing a porogen-containing dielectric material over the ILD layer, and then performing a curing process to drive out the porogen, thereby forming the dielectric layer **109** that is porous, as an example. Other suitable method may also be used to form the dielectric layers **109**.

As illustrated in FIG. **1**A, conductive features, such as conductive lines **105** and vias **107**, are formed in the dielectric layers **109**. In an example embodiment, the conductive features may include a diffusion barrier layer and a conductive material (e.g., copper, or a copper-containing material) over the diffusion barrier layer. The diffusion barrier layer may include titanium, titanium nitride, tantalum, tantalum nitride, or the like, and may be formed by CVD, Physical Vapor Deposition (PVD), Atomic Layer Deposition (ALD), or the like. After the diffusion barrier layer is formed, the conductive material is formed over the diffusion barrier layer. The formation of the conductive features may include a single damascene process, a dual damascene process, or the like.

Next, a plurality of passivation layers **111** are formed over the interconnect structures **106**, and a plurality of metal-insulator-metal (MIM) capacitors **113** are formed in the passivation layers **111**. The passivation layers **111** may be formed of one or more suitable dielectric materials such as silicon oxide, silicon nitride, low-k dielectrics such as carbon doped oxides, extremely low-k dielectrics such as porous carbon doped silicon dioxide, combinations of these, or the like. Each of the passivation layers **111** may be formed through a process such as chemical vapor deposition (CVD), FCVD, although any suitable process may be utilized.

The MIM capacitors **113** are formed in the passivation layers **111**. FIG. **1**B illustrates a zoomed-in view of an area **102** in FIG. **1**A to show details of the MIM capacitors **113**. As illustrated in FIG. **1**B, each of the MIM capacitors **113** includes two metal layers **113**M (e.g., copper layers) and a dielectric layer **113**I (e.g., a high-k dielectric layer) between the metal layers **113**M. Each of the layers (e.g., **113**M, **113**I, and **113**M) of the MIM capacitor **113** is formed in a respective passivation layer (e.g., **111**B, **111**C, or **111**D). The upper metal layer **113**M and the lower metal layer **113**M of the MIM capacitor **113** may be connected to an overlying via **119**V and an underlying via **108**, respectively, where the overlying via **119**V and the underlying via **108** are formed in passivation layers **111**E and **111**A, respectively, as an example. As another example, the upper metal layer **113**M and the lower metal layer **113**M of the MIM capacitor **113** may be connected to a first overlying via **119**V**1** and a second overlying via **119**V**2**, respectively. In the example of FIG. **1**B, the second overlying via **119**V**2** extends through the passivation layer **111**D and the dielectric layer **113**I to connect with the lower metal layer **113**M. Note that the

second overlying via **119**V**2** extends through an opening in the upper metal layer **113**M of the MIM capacitor, and therefore, is separated from (e.g., not contacting) the upper metal layer **113**M of the MIM capacitor by portions of the passivation layer **111**D.

Referring back to FIG. **1**A, the lower metal layer of the MIM capacitor **113** may be electrically coupled to a conductive feature of the interconnect structure **106**, e.g., through a via that extends from the lower metal layer of the MIM capacitor **113** to the conductive feature of the interconnect structure **106**. In addition, the plurality of MIM capacitors **113** may be electrically coupled in parallel to provide a large capacitance value. For example, the upper metal layers of the MIM capacitors **113** may be electrically coupled together, and the lower metal layers of the MIM capacitors **113** may be electrically coupled together. In some embodiments, the MIM capacitors **113** are omitted.

Referring next to FIG. **2**, openings **112** are formed in the passivation layer **111**. Some of the openings **112** extend through the passivation layer **111** to expose conductive features of the interconnect structure **106**. In some embodiments, some of the openings **112** extend partially through the passivation layer **111** to expose the upper metal layers of the MIM capacitors **113**. The openings **112** may be formed in one or more etching processes (e.g., anisotropic etching processes).

After the openings **112** are formed, a barrier layer **115** is formed conformally over the upper surfaces of the passivation layer **111** and along sidewalls and bottoms of the openings **112**. The barrier layer **115** may have a multi-layer structure and may include a diffusion barrier layer (e.g., a TiN layer) and a seed layer (e.g., a copper seed layer) formed over the diffusion barrier layer. The barrier layer **115** may be formed using any suitable formation method(s), such as CVD, PVD, ALD, combinations thereof, or the like.

Next, in FIG. **3**, a photoresist layer **137** is formed over the barrier layer **115**. The photoresist layer **137** is patterned (e.g., using photolithography technique) to form opening **138** at locations where conductive pads **119** and conductive lines **118** (see FIG. **4**) will be formed. The openings **138** expose, e.g., the seed layer of the barrier layer **115**. After the openings **138** are formed, a descum process **110** is performed to clean residues left by the patterning process of the photoresist layer **137**. The descum process **110** may be a plasma process performed using a process gas comprising oxygen, as an example.

Next, in FIG. **4**, conductive pads **119** (e.g., **119**A and **119**B) and conductive lines **118** are formed in the openings **138** over the barrier layer **115**. The conductive pads **119** and the conductive lines **118** may comprise an electrically conductive material, such as copper or copper alloy (e.g., a copper-silver alloy, a copper-cobalt alloy, or the like), and may be formed using a suitable formation method such as electroplating, electroless plating, or the like. After the conductive pads **119** and the conductive lines **118** are formed, the photoresist layer **137** is removed by a suitable removal process, such as ashing. Next, an etching process is performed to remove portions of the barrier layer **115** on which the conductive pads **119** or the conductive lines **118** are not formed. As illustrated in FIG. **4**, portions of the electrically conductive material fill the openings **112** (see FIG. **3**) in the passivation layer **111** to form vias **119**V, which electrically coupled the conductive pads **119** to underlying conductive features of the interconnect structure **106** or the MIM capacitor **113**. The conductive pads **119** and the conductive lines **118** may be collectively referred to as a redistribution layer (RDL), and the vias **119**V may be referred to as RDL vias. The shape of the cross-section of the conductive pad **119** may be a dome shape (e.g., with a curved upper surface), a concave shape, a polygon shape, or a rectangular (or square) shape, as examples. An area of the RDL via **119**V may be between about $0.9\times0.9\ \mu m^2$ and about $3.5\times3.5\ \mu m^2$, as an example.

Note that in FIG. **4**, some of the conductive pads **119** (e.g., **119**A) are larger (e.g., having a larger width measured between opposing sidewalls) than other conductive pads **119** (e.g., **119**B). In subsequent processing, C4 bumps (see, e.g., **125** in FIG. **8**A) will be formed on the larger conductive pads **119**A, and micro-bumps (see, e.g., **127** in FIG. **8**A) will be formed on the smaller conductive pads **119**B. In the illustrated embodiment, the conductive lines **118** are electrically coupled to conductive features of the interconnect structure **106**, e.g., through vias **116** (see FIG. **8**C), which vias **116** may not be in the cross-section of FIG. **4**. In other embodiments, the conductive lines **118** are dummy lines (e.g., electrically isolated). The number of conductive pads **119** and the number of conductive lines **118** may be any suitable number, and may be arranged in any order, as skilled artisans readily appreciate. In addition, the number of RDL vias **119**V under each of the conductive pad **119** may be any suitable number, such as one, two, three, or more. Furthermore, the RDL vias **119**V under each of the conductive pads **119** may be centered with respect to the conductive pad **119**, or may be off-center with respect to the conductive pad **119**.

Next, in FIG. **5**, a passivation layer **121** is conformally formed over the conductive pads **119**, over the conductive lines **118**, and over the passivation layer **111**. In some embodiments, the passivation layer **121** has a multi-layered structure and includes an oxide layer (e.g., silicon oxide) and a nitride layer (e.g., silicon nitride) over the oxide layer. In other embodiments, the passivation layer **121** has a single layer structure, e.g., having a single nitride layer. The passivation layer **121** may be formed using, e.g., CVD, PVD, ALD, combinations thereof, or the like.

Next, in FIG. **6**, a photoresist layer **135** is formed over the passivation layer **121** by, e.g., spin coating. The photoresist layer **135** is then patterned by, e.g., photolithography techniques to form openings **136** at locations where conductive bumps will be formed. Next, an etching process is performed to remove portions of the passivation layer **121** exposed by the openings **136** in the patterned photoresist layer **135**. In some embodiments, the etching process is a dry etch process (e.g., a plasma etching process) using a process gas comprising a mixture of CF_4, CHF_3, N_2, and Ar. Other process gas may also be used, e.g., O_2 may be used in place of CF_4. After the etching process, the conductive pads **119** are exposed. Next, the photoresist layer **135** is removed by a suitable removal process, such as ashing.

Next, in FIG. **7**, a dielectric layer **131** is formed over the passivation layer **121**, over the conductive pads **119**, and over the passivation layer **111**. Openings **132** are formed in the dielectric layer **131** to expose the underlying conductive pads **119**. The dielectric layer **131** may be formed of, e.g., polymer, polyimide (PI), benzocyclobutene (BCB), an oxide (e.g., silicon oxide), or a nitride (e.g., silicon nitride). The dielectric layer **131** is illustrated as a single layer in FIG. **7** as a non-limiting example. The dielectric layer **131** may have a multi-layer structure that includes a plurality of sub-layers formed of different materials.

In some embodiments, the dielectric layer **131** is a photosensitive material such as a photosensitive polymer material, and the openings **132** are formed by using photolithography techniques. For example, the photosensitive material

may be exposed to a patterned energy source (e.g., light) through, e.g., a reticle. The impact of the energy causes a chemical reaction in those parts of the photosensitive material that were impacted by the patterned energy source, thereby modifying the physical properties of the exposed portions of the photosensitive material such that the physical properties of the exposed portions of the photosensitive material are different from the physical properties of the unexposed portions of the photosensitive material. The photosensitive material may then be developed with a developer to remove the exposed portion of the photosensitive material or the unexposed portion of the photosensitive material, depending on, e.g., whether a negative photosensitive material or a positive photosensitive material is used. The remaining portions of the photosensitive material may be cured to form a patterned dielectric layer **131**. The top corners of the dielectric layer **131** at the openings **132**/**132**A are illustrated to be sharp (e.g., comprising two intersecting lines) in FIG. **7** as a non-limiting example. The top corners of the dielectric layer **131** at the openings **132**/**132**A may be, e.g., rounded corners.

In FIG. **7**, the larger opening **132**, denoted as opening **132**A, exposes the larger conductive pad **119**A (e.g., for forming C_4 bump), and the other smaller openings **132** expose the smaller conductive pads **119**B (e.g., for forming μ-bumps). Note that for each of the smaller openings **132**, a first distance between opposing sidewalls **131**S of the dielectric layer **131** exposed by the smaller opening **132** is larger than a second distance between opposing sidewalls **121**S of the passivation layer **121** exposed by the smaller opening **132**, such that the upper surfaces **121**U and the sidewalls **121**S of the passivation layer **121** are exposed by the smaller opening **132**. In other words, the upper portion of the smaller opening **132** is wider than the lower portion of the smaller opening **132**. The dielectric layer **131** is said to be pulled back from the sidewalls of the passivation layer **121** at the smaller openings **132**, and openings having the shape of the smaller openings **132** are referred to as pulled-back openings. In some embodiments, all of the smaller openings **132** for forming μ-bumps are pulled-back openings. Due to the smaller sizes of the μ-bumps and the conductive pads **119**B, it might be difficult to form the smaller openings **132** to be lined-up openings or pulled-in openings, details of the lined-up openings and the pulled-in openings are discussed hereinafter. On the other hand, due to the larger size of the C4 bumps and the conductive pads **119**A, the larger opening **132**A may be formed to be pulled-back openings, pulled-in openings, or lined-up openings.

In the example of FIG. **7**, the larger opening **132**A is a pulled-in opening, where the upper surfaces **121**U and the sidewalls **121**S of the passivation layer **121** underlying the larger opening **132**A are covered by the dielectric layer **131**. Therefore, the larger opening **132**A exposes the upper surface of the larger conductive pad **119**A and sidewalls of the dielectric layer **131**, but the sidewalls of the passivation layer **121** are not exposed by the larger opening **132**A. The width of the larger opening **132**A, measured between opposing sidewalls **131**S of the dielectric layer **131** facing the larger opening **132**A, is larger than a largest width of the smaller openings **132**, in some embodiments.

Next, in FIG. **8**A, conductive bumps **127** are formed on the smaller conductive pads **119**B, conductive bumps **125** are formed on the larger conductive pads **119**A, and dummy bumps **128** are formed over the conductive lines **118**. In an example embodiment, the conductive bumps **127** are μ-bumps having a width (e.g., measured between opposing sidewalls) between, e.g., about 3 μm and about 30 μm, and the conductive bumps **125** are C4 bumps having a width between about 32 μm and about 80 μm. In some embodiments, the dummy bump **128** has a same width as the conductive bump **127**. In other embodiments, the width of the dummy bump **128** is different from that of the conductive bump **127**, and is smaller than or same as the width of the conductive bump **125**.

The conductive bumps **127**/**125** and the dummy bumps **128** may be formed by forming a seed layer over the dielectric layer **131** and along sidewalls and bottoms of the openings **132**/**132**A; forming a patterned photoresist layer over the seed layer, where openings of the patterned photoresist layer are formed at locations where the conductive bumps **127**/**125** and the dummy bumps **128** are to be formed; forming (e.g., plating) an electrically conductive material over the seed layer in the openings; removing the patterned photoresist layer; and removing portions of the seed layer over which no conductive bumps **127**/**125** or no dummy bumps **128** are formed. Note that portions of the electrically conductive material fill the openings **132** and the openings **132**A to form bump vias **127**V and bump vias **125**V, respectively, which bumps vias **127**V/**125**V electrically couple the conductive bumps **127**/**125** to the underlying conductive pads **119**B/**119**A. The dummy bumps **128** are electrically isolated (e.g., not electrically coupled to other conductive features of the semiconductor device **100**). After the conductive bumps **127**/**125** and the dummy bumps **128** are formed, solder regions **129** may be formed on the upper surfaces of the conductive bumps **127**/**125** and the dummy bumps **128**.

The dummy bumps **128** may be formed to balance the bump density of the conductive bumps (e.g., the total area of the conductive bumps in a unit area, measured in a top view of the device) and to improve co-planarity of the conductive bumps (e.g., **125**, **127**, **128**) formed. If the bump densities of the conductive bumps in different areas of the semiconductor device **100** are not balanced, during the plating process to form the conductive bumps, the conductive bumps in areas with lower bump density may be formed taller than conductive bumps in areas with high bump density, thereby resulting in reduced co-planarity of the conductive bumps. Therefore, the dummy bumps **128** may be formed in areas of low bump density to even out the bump densities between different areas of the semiconductor device and to achieve improved co-planarity of the conductive bumps formed.

In FIG. **8**A, the dummy bump **128** is formed over three conductive lines **118** as a non-limiting example. The number of conductive lines **118** under each dummy bump **128** may be any suitable number, such as one. The number of bump vias **127**V or **125**V under a respective conductive bump (e.g., **127** or **125**) is one, as illustrated in FIG. **8**A. This is, of course, merely a non-limiting example. The number of bump vias **127**V or **125**V under a respective conductive bump (e.g., **127** or **125**) may be any suitable number, such as one, two, three, or more. In addition, the one or more bump vias **127**V or **125**V under each respective conductive bump (e.g., **127** or **125**) may be centered with respect to the conductive bump, or may be off-center with respect to the conductive bump. In some embodiments, when the dummy bump **128** is wider than the conductive lines **118**, the dummy bump **128** is formed over more than one conductive lines **118** to provide more support for the dummy bump **128** and more integration density for the conductive lines **118**.

Note that in FIG. **8**A, due to the difference in the shape of the opening **132** (e.g., a pulled-back opening) and the shape of the opening **132**A (e.g., a pulled-in opening), the bump via **127**V and the bump via **125**V have different shapes. For

example, the bump via **127**V has an upper portion (e.g., a portion above the upper surface of the passivation layer **121**) and a lower portion (e.g., a portion below the upper surface of the passivation layer **121**), where the upper portion is wider than the lower portion such that there is a lateral offset between a sidewall of the upper portion and a respective sidewall of the lower portion. In particular, sidewalls of the upper portion of the bump via **127**V contact and extend along sidewalls of the dielectric layer **131**, and sidewalls of the lower portion of the bump via **127**V contact and extend along sidewalls of the passivation layer **121**. As illustrated in FIG. **8**A, there is a step change in the width of the bump via **127**V between the upper portion and the lower portion of the bump via **127**V. In other words, there is an abrupt (e.g., non-continuous) change in the width of the bump via **127**V at the interface between the upper portion and the lower portion of the bump via **127**V.

In contrast, sidewalls of the bump via **125**V in FIG. **8**A contact and extend along sidewalls of the dielectric layer **131**. The width of the bump via **125**V may be constant (e.g., having sidewalls perpendicular to a major upper surface of the substrate **101**) or may change gradually (e.g., continuously) as the bump via **125**V extends toward the substrate **101**. In the example of the FIG. **8**A, the sidewalls of the bump via **125**V have a liner profile (e.g., a slanted straight line), and the width of the bump via **125**V decreases gradually as the bump via **125**V extends toward the substrate **101**. Note that there is a gap between the sidewall of the bump via **125**V and a respective sidewall of the passivation layer **121**, and the dielectric layer **131** fills the gap and contacts the upper surface of the conductive pads **119**A. In other words, the bump via **125**V is spaced apart (e.g., separated) from the passivation layer **121** by the dielectric layer **131**.

In the example of FIG. **8**A, a thickness A of a portion of the dielectric layer **131** disposed over the passivation layer **121** on the conductive line **118** is between about 2 μm and about 20 μm. A space S between the conductive line **118** and an adjacent conductive pad **119** is larger than about 1.5 μm if the conductive pad is the smaller conductive pad **119**B (e.g., with a p-bump formed thereon), or is larger than about 4 μm if the conductive pad is the larger conductive pad **119**A (e.g., with a C4 bump formed thereon). FIG. **8**A also illustrates a thickness T for sidewall portions of the passivation layer **121** (e.g., portions along sidewalls of the conductive pad **119** or along sidewalls of the conductive line **118**), and a thickness G for upper portions of the passivation layer **121** (e.g., portions along upper surfaces of the conductive pad **119** or along upper surfaces of the conductive line **118**), where the thickness G is between about 0.5 μm and about 5 μm, and where a ratio between T and G (e.g., T/G), referred to as the step coverage of the passivation layer **121**, is between about 30% and about 90%.

FIG. **8**B illustrates a zoomed-in view of a portion of the semiconductor device **100** of FIG. **8**A comprising a conductive bump **125**. Dimensions of the conductive bump **125** and its surrounding structures are discussed below. In addition to, or in parallel to, the discussion of the dimensions of the conductive bumps **125** (e.g., a C4 bump), the dimensions of the conductive bump **127** (e.g., a μ-bump) is also discussed where appropriate. Note that some features, such as the thickness of the passivation layer **121** or shape of the cross-section of the conductive pads **119**, are independent from the type of conductive bumps (e.g., C4 bump or μ-bump). In addition, some features, such as the width B at the top of the opening in the dielectric layer **131**, applies for embodiments where the dielectric layer **131** is formed.

As illustrated in FIG. **8**B, a width W of the conductive bump **125** (e.g., a C4 bump) is between about 32 μm and about 80 μm. In contrast, a width W of the conductive bump **127** (e.g., a μ-bump) is between about 10 μm and about 30 μm. A width B at the top of the opening in the dielectric layer **131** is between about 5 μm and about 25 μm for pulled-back openings for μ-bumps, and is between about 10 μm and about 80 μm for pulled-back openings, pulled-in openings, and lined-up openings for C4 bumps. A width E at the bottom of the opening in the dielectric layer **131** is between about 5 μm and about 30 μm for pulled-back openings for μ-bumps, and is between about 10 μm and about 80 μm for pulled-back openings, pulled-in openings, and lined-up openings for C4 bumps. A height D for the bump via **125**V (or **127**V) is larger than the thickness A (see FIG. **8**A) of a portion of the dielectric layer **131**, and is larger than the thickness G (see FIG. **8**A) of the upper portions of the passivation layer **121**. For embodiments where the dielectric layer **131** is omitted (see FIG. **15**), the height D is equal to the thickness G.

Still referring to FIG. **8**B, a width L of the conductive pad **119** is between about 15 μm and about 90 μm if the conductive pad is the larger conductive pad **119**A with C4 bump formed thereon, or is between about 3.5 μm and about 15 μm if the conductive pad is the smaller conductive pad **119**B with a μ-bump formed thereon. A ratio between the width L of the conductive pad **119** and the spacing S (see FIG. **8**A) is equal to or larger than one. A width E' of the opening in the passivation layer **121** is between about 2 μm and about 28 μm for μ-bumps, and is between about 30 μm and about 50 μm for C4 bumps. A height J of the conductive pad **119** (or of the conductive line **118**) is between about 2 μm and about 9 μm. FIG. **8**B further illustrates an angle F' between the sidewall of the dielectric layer **131** and the upper surface of the conductive pad **119**, and an angle F between the sidewall of the passivation layer **121** and the upper surface of the conductive pad **119**, where F may be between 5 degrees and 90 degrees (e.g., $5<F<90$), and F' is larger than 5 degrees and smaller than or equal to 90 degrees (e.g., $5<F'\geq 90$). If the openings **132** or **132**A is a lined-up opening (see **132**A in FIG. **13**), then the corresponding F and F' are equal. Otherwise, F may be different from F'.

FIG. **8**C illustrates a plan view of the semiconductor device **100** of FIG. **8**A, and FIG. **8**A corresponds to the cross-sectional view along cross-section A-A of FIG. **8**C. Note that for simplicity, not all features are illustrated in FIG. **8**C. The conductive pads **119** in FIG. **8**C are illustrated to have octagon shapes as a non-limiting example. Other shapes, such as circle shapes, oval shapes, rectangular shape, other polygon shapes, or the like, are also possible and are fully intended to be included within the scope of the current disclosure. As illustrated in FIG. **8**C, the dummy bump **128** overlies the conductive lines **118**, and the conductive lines **118** are electrically coupled to underlying conductive features (e.g., the interconnect structure **106**) through vias **116**.

FIG. **8**D illustrates an example of the arrangement of the conductive bumps (e.g., **125**, **127**) of the semiconductor device **100**. FIG. **8**D corresponds to a top view of a portion of the semiconductor device **100** of FIG. **8**A, which portion may be different from the portion illustrated in FIG. **8**A. Note that for simplicity, not all features are illustrated in FIG. **8**D. As illustrated in FIG. **8**D, conductive bumps **125** (e.g., C4 bumps) and conductive bumps **127** (e.g., μ-bumps) are interposed (e.g., interleaved) with each other in a same region of the semiconductor device **100**. For example, the conductive bumps **127** are disposed around and between the conductive bumps **125**. In addition, each of the conductive

bumps **125** is also interposed between the conductive bumps **127**. The layout of the conductive bumps (e.g., **125**, **127**) in FIG. **8**D differs from other designs where C4 bumps and μ-bumps are formed in different regions of the semiconductor device (e.g., not interposed between each other) and therefore, may also be referred to as a hybrid layout (or hybrid bump scheme) for the conductive bumps. In some embodiments, the conductive bumps **125** (e.g., C4 bumps) and conductive bumps **127** (e.g., μ-bumps) are formed in different groups in different regions of the semiconductor device (e.g., not interposed between each other).

The conductive bumps **127** in FIG. **8**D form a circle around each of the conductive bumps **125**. This is, of course, merely a non-limiting example. The conductive bumps **127** around each of the conductive bumps **125** may form any shape, such as a rectangular shape, a triangular shape, a circular shape, an oval shape, a polygon shape, as examples. In addition, the conductive bumps **127** around each conductive bump **125** may be formed in multiple rows and multiple columns. In some embodiments, a pitch P**1** between the conductive bumps **125** is between about 50 μm and about 180 μm, such as 75 μm, and a pitch P**2** between the conductive bumps **127** is between about 10 μm and about 45 μm, such as 25 μm.

Advantages are achieved for the disclosed semiconductor device **100** and other disclosed embodiments (e.g., **100**A, **100**B, **100**C) herein. For example, the dummy bump **128** is formed over conductive lines **118** instead of dummy conductive pads. This removes the need to form a dummy conductive pad under each of the dummy bump **128**, and allows for more flexible design and layout. For example, the space under the dummy bump **128** can now be used to route conductive lines **118**. As another example, the formation of the dielectric layer **131** removes some of the difficulties associated with forming the conductive bumps (e.g., **125**, **127**, **128**) and improves device reliability and manufacturing yield. Recall that to form the conductive bumps, a seed layer is formed first, then the conductive bumps are formed (e.g., plated) over the seed layer. Without the dielectric layer **131**, the seed layer would have to be formed conformally over the conductive pads **119** and over the conductive lines **118**. In advanced semiconductor manufacturing, the small gaps between the conductive pads **119** and the conductive lines **118** may have high aspect-ratios, and it may be difficult to form the seed layer in these small gaps, which may result in the conductive bumps not being formed properly. In addition, after the conductive bumps are formed, portions of the seed layer over which no conductive bump is formed need to be removed. If such portions of the seed layer are in the small gaps, it may be difficult to remove the seed layer, which may result in electrical short between conductive pads **119**. In contrast, with the dielectric layer **131** formed, the seed layer is formed over the dielectric layer **131** and in the openings **132**/**132**A, which openings **132**/**132**A have much smaller aspect-ratios, and therefore, the seed layer can be easily formed in the openings **132** and easily removed from the openings **132**. Furthermore, the hybrid layout allows for more flexible placement of the conductive bumps in the design to accommodate different design requirements.

FIGS. **9** and **10** illustrate cross-sectional views of a semiconductor device **100**A at various stages of manufacturing, in accordance with another embodiment. The semiconductor device **100**A is similar to the semiconductor device **100**, but with pulled-back openings **132**A for the conductive pads **119**A. The processing of FIG. **9** follows the processing of FIGS. **1**A, **1**B, and **2**-**6**. In FIG. **9**, the dielectric layer **131** is formed, and openings **132**/**132**A are formed, using the same or similar processing as discussed above with reference to FIG. **7**. Note that in FIG. **9**, the opening **132**A is formed as a pulled-back opening. This may be achieved by, e.g., using a reticle with a larger opening at the location of the opening **132**A during the photolithography process to form the opening **132**A. Therefore, in the example of FIG. **9**, all the openings **132**/**132**A are pulled-back openings.

Next, in FIG. **10**, conductive bumps **125** (e.g., C4 bumps) are formed over the larger conductive pads **119**A, conductive bumps **127** (e.g., μ-bumps) are formed over the smaller conductive pads **119**B, and dummy bumps **128** are formed over the conductive lines **118**, following the same or similar processing as FIG. **8**A, details are not repeated. In the example of FIG. **10**, the bumps vias **125**V and **127**V have similar shapes (e.g., having upper portions and lower portions with different widths, and having a step change in the width at the interface between the upper portion and the lower portion). Details are the same as or similar to those of the bump vias **127**V in FIG. **8**A, thus not repeated.

FIGS. **11**-**14** illustrate cross-sectional views of a semiconductor device **100**B at various stages of manufacturing, in accordance with another embodiment. The semiconductor device **100**B is similar to the semiconductor device **100**, but with lined-up openings **132**A for the conductive pads **119**A.

The processing of FIG. **11** follows the processing of FIGS. **1**A, **1**B, and **2**-**5**. In FIG. **11**, the patterned photoresist layer **135** is formed, and an etching process is performed to form openings **136** that extend through the passivation layer **121** to expose the smaller conductive pads **119**B, using the same or similar processing as FIG. **6**. Note that unlike the processing of FIG. **6**, no opening is formed over the larger conductive pad **119**A. Therefore, the larger conductive pad **119**A is covered by the photoresist layer **135**. The photoresist layer **135** is removed after the openings **136** are formed.

Next, in FIG. **12**, the dielectric layer **131** is formed, and openings **132** are formed to expose the underlying smaller conductive pads **119**B, using the same or similar processing as FIG. **7**. Note that unlike the processing of FIG. **7**, no opening is formed over the larger conductive pad **119**A in the dielectric layer **131**.

Next, in FIG. **13**, a photoresist layer **133** is formed over the dielectric layer **131**. The photoresist layer **133** fills the openings **132**. Next, the photoresist layer **133** is patterned to form openings **132**A over the larger conductive pads **119**A. The patterned photoresist layer **133** is then used as an etching mask for a subsequent etching process, which subsequent etching process may be the same as or similar to the etching process in FIG. **6** for forming the openings **136**. As illustrated in FIG. **13**, after the etching process, openings **132**A, which are lined-up openings, are formed over and expose the underlying larger conductive pads **119**A. For each lined-up opening **132**A, the sidewall **131**S of the dielectric layer **131** exposed by the opening and a respective sidewall **121**S of the passivation layer **121** exposed by the opening are aligned along a same line (e.g., a straight line or a slanted line with respect to a major upper surface of the substrate **101**), as illustrated in the example of FIG. **13**. In some embodiments, the sidewall **131**S of the dielectric layer **131** and the respective sidewall **121**S of the passivation layer **121** may intersect at the interface between the dielectric layer **131** and the passivation layer **121** without a lateral offset in between. The photoresist layer **133** is removed after the lined-up openings **132**A are formed.

Next, in FIG. **14**, conductive bumps **125** (e.g., C4 bumps) are formed over the larger conductive pads **119**A, conductive bumps **127** (e.g., μ-bumps) are formed over the smaller

conductive pads **119**B, and dummy bumps **128** are formed over the conductive lines **118**, following the same or similar processing as FIG. **8**A, details are not repeated. In the example of FIG. **14**, the bump via **127**V of the conductive bump **127** (e.g., μ-bump) has the same shape as the bump via **127**V of FIG. **8**A. The sidewalls of the bump via **125**V contact and extend along sidewalls **131**S of the dielectric layer **131**, and contact and extend along sidewalls **121**S of the passivation layer **121**. In some embodiments, the bump via **125**V of the conductive bump **125** (e.g., C4 bump) has a width (e.g., measured between opposing sidewalls of the bump via **125**V) that is constant (e.g., having straight sidewalls) or changes continuously (e.g., gradually without a step change) as the bump via **125**V extends toward the substrate.

FIG. **15** illustrates a cross-sectional view of a semiconductor device **100**C, in accordance with yet another embodiment. The semiconductor device **100**C is similar to the semiconductor device **100** of FIG. **8**A, but without the dielectric layer **131**. As illustrated in FIG. **15**, the conductive bumps **125**/**127** and the dummy bumps **128** are formed directly on (e.g., in physical contact with) the passivation layer **121**. Bump vias **125**V and **127**V extend through the passivation layer **121**, and electrically couple the conductive bumps **125** and **127** to the underlying conductive pads **119**A and **119**B, respectively. Note that the dummy bump **128** is electrically isolated from the conductive lines **118** by the passivation layer **121**. Lower portions of the dummy bumps **128** extend into the gaps between the conductive lines **118** in the example of FIG. **15**, and therefore, are closer to the substrate **101** than the conductive bumps **125**/**127**.

FIGS. **16**A and **16**B illustrate cross-sectional views of a semiconductor device **200** with different processing sequences, in accordance with some embodiments. The semiconductor device **200** may be formed by a same or similar process as the semiconductor device **100**A of FIG. **10**. FIGS. **16**A and **16**B illustrate the effect of the sequence of the processing steps on the angles F and F', where the angle F is between the sidewall **121**S of the passivation layer **121** and the underlying upper surface of the conductive pad **119**, and the angle F' is between the sidewall **131**S of the dielectric layer **131** and the underlying upper surface of the passivation layer **121**.

For the semiconductor device **200** in FIG. **16**A, after the passivation layer **121** is formed over the conductive pad **119**, the dielectric layer **131** is formed over the passivation layer **121**. In an embodiment, the dielectric layer **131** is a photosensitive polymer layer and is formed by spin coating. Next, the dielectric layer **131** (e.g., a photosensitive polymer layer) is exposed to a patterned energy source (e.g., light) and developed to form a patterned dielectric layer **131**. The patterned dielectric layer **131** is used as an etching mask in a subsequent etching process, which subsequent etching process etches through the passivation layer **121** to form openings that expose the conductive pads **119**. After the openings in the passivation layer **121** are formed, the dielectric layer **131** (e.g., a photosensitive polymer layer) is cured, e.g., by a thermal process. In other words, curing of the dielectric layer **131** is performed after the openings in the passivation layer **121** are formed. Next, the conductive bumps **125** or **127** are formed, following similar processing as discussed above.

Note that in FIG. **16**A, by curing the dielectric layer **131** after the openings in the passivation layer **121** are formed, the angle F' is larger than 5 degrees and equal to or smaller than 90 degrees (e.g., $5° < F' \leq 90°$), and the angle F is larger than 5 degrees and smaller than 90 degrees (e.g., $5° < F < 90°$).

The semiconductor device **200** in FIG. **16**B is formed by the similar processing as FIG. **16**A, but with the curing of the dielectric layer **131** performed before the openings in the passivation layer **121** are formed. In particular, after the dielectric layer **131** (e.g., a photosensitive polymer layer) is exposed and developed, the dielectric layer **131** is cured, e.g., by a thermal process. Next, the cured patterned dielectric layer **131** is used as an etching mask in a subsequent etching process, which subsequent etching process etches through the passivation layer **121** to form openings that expose the conductive pads **119**. Next, the conductive bumps **125** or **127** are formed, following similar processing as discussed above.

Note that in FIG. **16**B, by curing the dielectric layer **131** before the openings in the passivation layer **121** are formed, the angle F' is larger than 5 degrees and smaller than 90 degrees (e.g., $5° < F' < 90°$), and the angle F is larger than 5 degrees and smaller than 90 degrees (e.g., $5° < F < 90°$). In some embodiments, the processing sequence of FIG. **16**A (e.g., curing the dielectric layer **131** after the openings in the passivation layer **121** are formed) allows for a larger angle F' (e.g., steeper sidewalls of the dielectric layer **131**) to be formed than the processing sequences in FIG. **16**B (e.g., curing the dielectric layer **131** before the openings in the passivation layer **121** are formed). The larger angle F' may advantageously reduce the stress in the device formed, e.g., at the corner of the dielectric layer **131**, and may prevent or reduce cracking or peeling of the materials (e.g., **131**) of the semiconductor device.

Embodiments of the present disclosure achieve some advantageous features. For example, the dummy bumps **128** are formed over conductive lines **118** instead of dummy conductive pads. This allows for more flexible design and layout. As another example, the formation of the dielectric layer **131** removes the difficulties associated with forming and removing the seed layer in high aspect-ratio gaps during formation of the conductive bumps (e.g., **125**, **127**, **128**), thereby improving device reliability and product yield. Furthermore, the hybrid layout allows for more flexible placement of the conductive bumps in the design to accommodate different design requirements.

FIG. **17** illustrates a flow chart of a method of fabricating a semiconductor structure, in accordance with some embodiments. It should be understood that the embodiment method shown in FIG. **17** is merely an example of many possible embodiment methods. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. For example, various steps as illustrated in FIG. **17** may be added, removed, replaced, rearranged, or repeated.

Referring to FIG. **17**, at block **1010**, an interconnect structure is formed over a substrate. At block **1020**, a first conductive pad, a second conductive pad, and a conductive line are formed over and electrically coupled to the interconnect structure, wherein the second conductive pad has a width larger than that of the first conductive pad. At block **1030**, a passivation layer is formed conformally over the first conductive pad, the second conductive pad, and the conductive line. At block **1040**, a first opening and a second opening are formed in the passivation layer to expose the first conductive pad and the second conductive pad, respectively, while keeping the conductive line covered by the passivation layer. At block **1050**, a first conductive bump, a second conductive bump, and a dummy bump are formed over the first conductive pad, the second conductive pad, and the conductive line, respectively. At block **1060**, a first bump via is formed under the first conductive bump and a second bump via is formed under the second conductive bump,

wherein the first bump via and the second bump via extend through the passivation layer, and electrically couple the first conductive bump and the second conductive bump to the first conductive pad and the second conductive pad, respectively, wherein the dummy bump is electrically isolated.

In accordance with an embodiment of the present disclosure, a semiconductor device includes a substrate; an interconnect structure over the substrate; a first passivation layer over the interconnect structure; a first conductive pad, a second conductive pad, and a conductive line that are disposed over the first passivation layer and electrically coupled to conductive features of the interconnect structure; a conformal second passivation layer over and extending along upper surfaces and sidewalls of the first conductive pad, the second conductive pad, and the conductive line; a first conductive bump and a second conductive bump over the first conductive pad and the second conductive pad, respectively, wherein the first conductive bump and the second conductive bump extend through the conformal second passivation layer and are electrically coupled to the first conductive pad and the second conductive pad, respectively; and a dummy bump over the conductive line, wherein the dummy bump is separated from the conductive line by the conformal second passivation layer. In an embodiment, the dummy bump comprises a same electrically conductive material as the first conductive bump and the second conductive bump, wherein the dummy bump is electrically isolated from the conductive line and the interconnect structure. In an embodiment, the semiconductor device further comprises a dielectric layer between the conformal second passivation layer and the dummy bump, wherein a lower surface of the dummy bump facing the substrate contacts and extends along an upper surface of the dielectric layer distal from the substrate. In an embodiment, the semiconductor device further comprises a first bump via and a second bump via, wherein the first bump via extends through the dielectric layer and the conformal second passivation layer to electrically couple the first conductive bump to the first conductive pad, wherein the second bump via extends through the dielectric layer and the conformal second passivation layer to electrically couple the second conductive bump to the second conductive pad. In an embodiment, a first width of the first conductive bump is smaller than a second width of the second conductive bump. In an embodiment, a third width of the dummy bump is a same as the first width. In an embodiment, the first bump via extends from the upper surface of the dielectric layer to the first conductive pad, wherein a width of the first bump via has a step change. In an embodiment, the second bump via extends from the upper surface of the dielectric layer to the second conductive pad, wherein the dielectric layer is disposed laterally between a sidewall of the second bump via and a sidewall of the conformal second passivation layer facing the second bump via. In an embodiment, the second bump via extends from the upper surface of the dielectric layer to the second conductive pad, wherein a width of the second bump via has a step change. In an embodiment, the second bump via extends from the upper surface of the dielectric layer to the second conductive pad, wherein a width of the second bump via changes continuously as the second bump via extends toward the substrate. In an embodiment, the semiconductor device further comprises a metal-insulator-metal (MIM) capacitor embedded in the first passivation layer. In an embodiment, in a top view, the first conductive bump is smaller than the second conductive bump, wherein the semiconductor device further comprises a third conductive bump having a same size as the second conductive bump, wherein the first conductive bump is disposed between the second conductive bump and the third conductive bump.

In accordance with an embodiment of the present disclosure, a semiconductor device includes an interconnect structure over a substrate; a first conductive pad, a second conductive pad, and a conductive line over and electrically coupled to the interconnect structure; a passivation layer over the first conductive pad, the second conductive pad, and the conductive line, wherein the passivation layer is conformal and extends along exterior surfaces of the first conductive pad, the second conductive pad, and the conductive line; a first conductive bump, a second conductive bump, and a dummy bump over the first conductive pad, the second conductive pad, and the conductive line, respectively, wherein a first width of the first conductive bump is smaller than a second width of the second conductive bump; and a first bump via and a second bump via under the first conductive bump and the second conductive bump, respectively, wherein the first bump via extends through the passivation layer and contacts the first conductive pad, wherein the second bump via extends through the passivation layer and contacts the second conductive pad, wherein the dummy bump is separated from the conductive line by the passivation layer and is electrically isolated from the conductive line. In an embodiment, the semiconductor device further comprises a dielectric layer over the passivation layer, wherein the first bump via extends into the dielectric layer and contacts a first upper surface of the first conductive pad, and wherein the second bump via extends into the dielectric layer and contacts a second upper surface of the second conductive pad. In an embodiment, the first bump via has a first sidewall contacting the dielectric layer and has a second sidewall contacting the passivation layer, wherein there is a first lateral offset between the first sidewall and the second sidewall. In an embodiment, the second bump via has a third sidewall contacting the dielectric layer and has a fourth sidewall contacting the passivation layer, wherein there is a second lateral offset between the third sidewall and the fourth sidewall. In an embodiment, sidewalls of the second bump via contact and extend along the dielectric layer and the passivation layer, wherein a distance between the sidewalls of the second bump via changes continuously as the second bump via extends toward the substrate. In an embodiment, the second bump via is spaced apart from the passivation layer.

In accordance with an embodiment of the present disclosure, a method of forming a semiconductor device includes forming an interconnect structure over a substrate; forming a first conductive pad, a second conductive pad, and a conductive line over and electrically coupled to the interconnect structure, wherein the second conductive pad has a width larger than that of the first conductive pad; conformally forming a passivation layer over the first conductive pad, the second conductive pad, and the conductive line; forming a first opening and a second opening in the passivation layer to expose the first conductive pad and the second conductive pad, respectively, while keeping the conductive line covered by the passivation layer; forming a first conductive bump, a second conductive bump, and a dummy bump over the first conductive pad, the second conductive pad, and the conductive line, respectively; and forming a first bump via under the first conductive bump and forming a second bump via under the second conductive bump, wherein the first bump via and the second bump via extend through the passivation layer, and electrically couple the first conductive bump and the second conductive bump

to the first conductive pad and the second conductive pad, respectively, wherein the dummy bump is electrically isolated. In an embodiment, the method further comprises after forming the first opening and the second opening and before forming the first conductive bump, the second conductive bump, and the dummy bump: forming a dielectric layer over the passivation layer, wherein the dielectric layer fills the first opening and the second opening; and forming a third opening and a fourth opening in the dielectric layer over locations of the first opening and the second opening to expose the first conductive pad and the second conductive pad, respectively, wherein the first bump via and the second bump via are formed in the third opening and the fourth opening, respectively, wherein after the dummy bump is formed, the dummy bump is separated from the conductive line by the passivation layer and the dielectric layer.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor device comprising:

a substrate;

electrical components over the substrate, wherein the electrical components comprise a transistor;

an interconnect structure over the substrate, wherein the electrical components are interconnected by the interconnect structure and form a functional circuit of a functional die;

a first passivation layer over the interconnect structure;

a metal-insulator-metal (MIM) capacitor embedded in the first passivation layer;

a first conductive pad, a second conductive pad, and a conductive line that are disposed over the first passivation layer and electrically coupled to respective conductive features of the interconnect structure;

a first via under the second conductive pad, wherein the first via extends into the first passivation layer and electrically couples the second conductive pad to the MIM capacitor, wherein the first via extends through an upper metal layer of the MIM capacitor without contacting the upper metal layer, wherein the first via contacts an upper surface of a lower metal layer of the MIM capacitor;

a conformal second passivation layer over and extending along upper surfaces and sidewalls of the first conductive pad, the second conductive pad, and the conductive line;

a first conductive bump and a second conductive bump over and electrically coupled to the first conductive pad and the second conductive pad, respectively;

a first bump via and a second bump via under the first conductive bump and the second conductive bump, respectively, wherein the first bump via extends through the conformal second passivation layer to electrically couple the first conductive bump to the first conductive pad, wherein the second bump via extends through the conformal second passivation layer to electrically couple the second conductive bump to the second conductive pad, wherein a first sidewall profile of the first bump via is different from a second sidewall profile of the second bump via; and

a dummy bump over the conductive line, wherein the dummy bump is separated from the conductive line by the conformal second passivation layer, wherein the first conductive bump, the second conductive bump, and the dummy bump are external connectors of the functional die.

2. The semiconductor device of claim **1**, wherein a first width of the first conductive bump, measured between sidewalls of the first conductive bump, is smaller than a second width of the second conductive bump measured between sidewalls of the second conductive bump.

3. The semiconductor device of claim **2**, further comprising a dielectric layer over the conformal second passivation layer, wherein the first conductive bump, the second conductive bump, and the dummy bump are disposed over an upper surface of the dielectric layer distal from the substrate, wherein the first bump via and the second bump via are disposed below the upper surface of the dielectric layer, wherein a lowermost surface of the dummy bump facing the substrate contacts and extends along the upper surface of the dielectric layer.

4. The semiconductor device of claim **3**, wherein an upper portion of the first bump via contacts and extends along the dielectric layer, and a lower portion of the first bump via contacts and extends along the conformal second passivation layer, wherein at an interface between the upper portion and the lower portion of the first bump via, there is a step change between a width of the upper portion of the first bump via and a width of the lower portion of the first bump via.

5. The semiconductor device of claim **4**, wherein the width of the upper portion of the first bump via is larger than the width of the lower portion of the first bump via.

6. The semiconductor device of claim **4**, wherein a width of the second bump via changes continuously as the second bump via extends from the upper surface of the dielectric layer toward the substrate.

7. The semiconductor device of claim **6**, wherein sidewalls of the second bump via extend along slanted straight lines.

8. The semiconductor device of claim **1**, wherein the dummy bump, the first conductive bump, and the second conductive bump comprise a same electrically conductive material, wherein the dummy bump is electrically isolated.

9. The semiconductor device of claim **3**, wherein there is no electrically conductive feature in the dielectric layer between the dummy bump and the conductive line.

10. A semiconductor device comprising:

a substrate;

electrical components over the substrate, wherein the electrical components comprise a transistor;

an interconnect structure over the substrate, wherein the electrical components are interconnected by the interconnect structure and form a functional circuit of a functional die;

a first passivation layer over the interconnect structure;

a first conductive pad, a second conductive pad, and a conductive line that are over the first passivation layer and electrically coupled to the interconnect structure;

a first via and a second via that are disposed under, and in contact with, a lower surface of the second conductive pad, wherein the lower surface of the second conductive pad extends along the first passivation layer, wherein the first via and the second via extend from the

lower surface of the second conductive pad toward the first passivation layer, wherein the first via extends closer to the substrate than the second via;

a second passivation layer over the first conductive pad, the second conductive pad, and the conductive line, wherein the second passivation layer is conformal and extends along exterior surfaces of the first conductive pad, the second conductive pad, and the conductive line;

a first conductive bump, a second conductive bump, and a dummy bump over the first conductive pad, the second conductive pad, and the conductive line, respectively, wherein the first conductive bump is narrower than the second conductive bump; and

a first bump via and a second bump via under the first conductive bump and the second conductive bump, respectively, wherein the first bump via extends through the second passivation layer and contacts the first conductive pad, wherein the second bump via extends through the second passivation layer and contacts the second conductive pad, wherein the first bump via is narrower than the second bump via, wherein the dummy bump is separated from the conductive line by the second passivation layer and is electrically isolated, wherein the first conductive bump, the second conductive bump, and the dummy bump are external connectors of the functional die.

11. The semiconductor device of claim **10**, further comprising a dielectric layer over the second passivation layer, wherein the first bump via extends into the dielectric layer and contacts a first upper surface of the first conductive pad, wherein the second bump via extends into the dielectric layer and contacts a second upper surface of the second conductive pad, wherein a lowermost surface of the dummy bump facing the substrate contacts and extends along an upper surface of the dielectric layer distal from the substrate, wherein there is no electrically conductive feature in the dielectric layer between the dummy bump and the conductive line.

12. The semiconductor device of claim **11**, wherein an upper portion of the first bump via has a first sidewall contacting the dielectric layer, and a lower portion of the first bump via has a second sidewall contacting the second passivation layer, wherein there is a lateral offset between the first sidewall and the second sidewall.

13. The semiconductor device of claim **12**, wherein an upper portion of the second bump via has a third sidewall contacting the dielectric layer, and a lower portion of the second bump via has a fourth sidewall contacting the second passivation layer, wherein there is no lateral offset between the third sidewall and the fourth sidewall.

14. The semiconductor device of claim **11**, wherein a distance between opposing sidewalls of the first bump via has a step change at an interface between the dielectric layer and the second passivation layer, wherein a distance between opposing sidewalls of the second bump via changes continuously as the second bump via extends toward the substrate.

15. The semiconductor device of claim **14**, wherein a first upper surface of a first portion of the second passivation layer at the first upper surface of the first conductive pad is exposed by the dielectric layer, wherein a second upper surface of a second portion of the second passivation layer at the second upper surface of the second conductive pad is covered by the dielectric layer.

16. The semiconductor device of claim **10**, further comprising a metal-insulator-metal (MIM) capacitor embedded in the first passivation layer, wherein the first via extends through the first passivation layer and is electrically coupled to a conductive feature of the interconnect structure, wherein the second via is electrically coupled to an electrode of the MIM capacitor.

17. The semiconductor device of claim **10**, wherein the first conductive bump, the second conductive bump, and the dummy bump comprise a same conductive material.

18. The semiconductor device of claim **14**, wherein the opposing sidewalls of the second bump via are slanted sidewalls.

19. A semiconductor device comprising:

a substrate;

electrical components over the substrate, wherein the electrical components comprise a transistor;

an interconnect structure over the substrate, wherein the electrical components are interconnected by the interconnect structure and form a functional circuit of a functional die;

a first passivation layer over the interconnect structure;

a metal-insulator-metal (MIM) capacitor embedded in the first passivation layer;

a first conductive pad, a second conductive pad, and a conductive line over the first passivation layer and electrically coupled to the interconnect structure, wherein the second conductive pad has a width larger than that of the first conductive pad;

a first via and a second via that are disposed under and connected with the second conductive pad, wherein the first via extends closer to the substrate than the second via, wherein the first via extends through the first passivation layer and is electrically coupled to a conductive feature of the interconnect structure, wherein the second via is electrically coupled to an electrode of the MIM capacitor;

a conformal second passivation layer extending along exterior surfaces of the first conductive pad, the second conductive pad, and the conductive line;

a dielectric layer over the conformal second passivation layer;

a first conductive bump, a second conductive bump, and a dummy bump over the dielectric layer;

a first bump via under the first conductive bump, wherein the first bump via extends through the dielectric layer and the conformal second passivation layer, and electrically couples the first conductive bump to the first conductive pad, wherein there is a lateral offset between a first sidewall of an upper portion of the first bump via and a respective second sidewall of a lower portion of the first bump via; and

a second bump via under the second conductive bump, wherein the second bump via extends through the dielectric layer and the conformal second passivation layer, and electrically couples the second conductive bump to the second conductive pad, wherein a first sidewall of an upper portion of the second bump via and a respective second sidewall of a lower portion of the second bump via extend along a same line, wherein the dummy bump is separated from the conductive line by the conformal second passivation layer and the dielectric layer, and is electrically isolated, wherein the first conductive bump, the second conductive bump, and the dummy bump are external connectors of the functional die.

20. The semiconductor device of claim **19**, wherein the upper portion of the first bump via and the upper portion of the second bump via extend along a first sidewall of the

dielectric layer and a second sidewall of the dielectric layer, respectively, wherein the lower portion of the first bump via and the lower portion of the second bump via extend along a first sidewall of the conformal second passivation layer and a second sidewall of the conformal second passivation layer, respectively.

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